

POSTERIOR VARIANCE COLLAPSE IN CONDITIONAL FLOW MATCHING IS THE ZERO-BANDWIDTH LIMIT OF A KERNEL-WEIGHTED EMPIRICAL CONDITIONAL FLOW

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ABSTRACT

Conditional flow matching (CFM) trained to convergence on a finite dataset is often described as “memorising” the training data: overtraining is reported to collapse the conditional generative law onto individual training examples. We give an exact, population-level account of why this happens. Fixing the dataset and the deterministic linear interpolant, the L^2 -optimal velocity field is a posterior-weighted average of per-example collapsed fields, and under hard conditioning with identifying labels this average degenerates to a single atom: the exact minimiser has *zero* conditional variance and zero loss (Proposition 2). We then show that the standard remedy—label smoothing, i.e. adding Gaussian noise to the conditioning variable—turns this hard selection into *exact kernel regression over the training atoms*: the generated endpoint law becomes a Nadaraya–Watson mixture $\sum_i p_i^{(h)}(y) \delta_{x^i}$ (Theorem 1) whose moments are a parameter-free reference curve computable directly from the training set (Proposition 5). This remedy, however, can only match moments; it cannot recover the posterior, since the generated law remains atomic at every bandwidth h , yielding an h -independent Wasserstein floor (Proposition 6). That obstruction names its own cure—move the support—and smoothing the *endpoint* does so, restoring an absolutely continuous law (Proposition 7); measuring the floor directly shows it is small at our default configuration, correcting an earlier reading of our own MMD curves. Two superficially similar remedies are moreover not interchangeable: label noise moves the population endpoint, interpolant noise provably does not, at any noise level (Proposition 9). Across five seeds, a trained MLP tracks the kernel reference to within seed noise (ratio-to-kernel ≈ 1.00 for $h \geq 0.05$), and we reproduce the same collapse on a Gaussian-mixture posterior, on MNIST and CIFAR-10 inpainting, and under adversarially shuffled labels. Throughout, we distinguish carefully between exact population consequences and empirical findings that the population theory does not, by itself, determine.

1 INTRODUCTION

Flow matching and stochastic-interpolant models (Lipman et al., 2023; Albergo and Vanden-Eijnden, 2023; Liu et al., 2023; Albergo et al., 2023) are usually trained by regressing a velocity field against a linear path between a noise sample and a data sample (Tong et al., 2024). On a finite dataset, exact minimisation of this objective against the *empirical* data law is known to induce memorisation: deployed diffusion models can be made to reproduce individual training images verbatim (Carlini et al., 2023; Somepalli et al., 2023), and a recent line of work has documented that conditional flow-matching models, trained past the point of good posterior calibration, produce increasingly deterministic (low-variance) conditional samples that coincide with individual training examples (Dasgupta et al., 2026; Reu et al., 2025).

We give an exact, population-level account of this mechanism. Fixing the training set $\mathcal{D} = \{(x^i, y^i)\}_{i=1}^N$ and treating only the interpolation randomness as random, we characterise the ex-

act population minimiser of the flow-matching objective over all square-integrable vector fields, together with the endpoint law of its flow.

That the empirical minimiser memorises is *not* new; it is well established in the diffusion-memorisation literature (Pidstrigach, 2022; Gu et al., 2023; Kadkhodaie et al., 2024; Biroli and Mézard, 2023; Kamb and Ganguli, 2024; Buchanan et al., 2025). Closest in spirit is Buchanan et al. (2025), who characterise a memorisation/generalisation *phase transition* in *unconditional* diffusion models as denoiser capacity M ranges from the number of modes K up to the dataset size N . That is a capacity-driven account of the same broad phenomenon we study from the label/bandwidth side—their $M: K \rightarrow N$ interpolation is the capacity analogue of our $h: 0 \rightarrow \infty$ interpolation (Corollaries 2 and 3)—and neither result subsumes the other. Our contribution is the *conditional* and *kernel-regression structure* underlying the collapse, which this literature does not isolate:

1. **Conditional sharpening.** Hard conditioning turns full-empirical-measure memorisation into *single-atom* memorisation—zero conditional variance (Corollary 1). The correct contrast with the unconditional model is not “memorises vs. does not” but *which* empirical measure is memorised. This is not the ordinary finite-sample memorisation of an over-parameterised network fitting a fixed batch (Appendix B, adapted from Reu et al., 2025, is exactly that phenomenon, for the *unconditional* model): under hard conditioning y^i identifies the training atom *deterministically*, so the collapse is already present in the exact population minimiser, for every x_0 and every fresh draw—no finite batch, no fitting, no resampling escape route.
2. **Label smoothing $\equiv$ kernel regression.** Label noise makes the population endpoint an exact Nadaraya–Watson mixture (Nadaraya, 1964; Watson, 1964) over training atoms (Theorem 1), converting a 0/1 statement into a quantitative, parameter-free reference curve $\text{Cov}_h(y)$ (Proposition 5).
3. **Separation of two remedies.** Label noise changes the population endpoint; interpolant noise provably does not, at any noise level (Propositions 9 and 10).
4. **Atomicity obstruction, and the escape from it.** The smoothed law is atomic for every h , yielding an h -independent Wasserstein floor (Proposition 6). The obstruction names its own cure—move the support—and endpoint smoothing does exactly that (Proposition 7). We *measure* the floor rather than assert it, and find it small at our default (d, N) , correcting an earlier reading of our own MMD curves.
5. **Representation vs. optimisation.** A uniform Lipschitz bound gives a measurable loss floor $d/(3L)$ (Corollary 4); measuring it shows the observed plateau is *not* explained by that mechanism. Separately (Appendix B) no finite network with globally Lipschitz activations represents the collapsed field’s $1/(1-t)$ blow-up pointwise on $[0, 1]$ —a claim we keep explicitly distinct from a loss floor.

In one line: *conditional collapse is the zero-bandwidth limit of a kernel-weighted empirical conditional flow*. On scope (Section 3.7): the theory characterises the exact population minimiser for the empirical data law, not what a finite network trained by SGD attains, so we separate quantities it determines from quantities it does not—the dependence of observed collapse on N or σ_{obs} —which we report as empirical findings, never as “predictions.”

2 SETUP AND NOTATION

Fix a training set $\mathcal{D} = \{(x^i, y^i)\}_{i=1}^N$ with $x^i \in \mathbb{R}^d$, $y^i \in \mathbb{R}^k$, empirical joint law $\hat{\rho}_{XY} = \frac{1}{N} \sum_i \delta_{(x^i, y^i)}$ and marginal $\hat{\rho}_X = \frac{1}{N} \sum_i \delta_{x^i}$. The training set is fixed (deterministic); randomness comes only from

$$I \sim \text{Unif}\{1, \dots, N\}, \quad X_0 \sim \pi_0 = \mathcal{N}(0, I_d), \quad t \sim \text{Unif}(0, 1),$$

drawn independently, together with label noise $\varepsilon \sim \mathcal{N}(0, I_k)$ when $h > 0$. Set $X_1 = x^I$ and, for the deterministic linear interpolant,

$$X_t = (1-t)X_0 + tX_1, \quad U := \dot{X}_t = X_1 - X_0.$$

The smoothed label is $\tilde{Y} = y^I + h\varepsilon$ with Gaussian kernel $K_h(u) = (2\pi h^2)^{-k/2} \exp(-|u|^2/2h^2)$. The objective is

$$L_h(v) = \mathbb{E}[|U - v(X_t, t, \tilde{Y})|^2]. \quad (1)$$

For $h = 0$ we write $\tilde{Y} = Y = y^I$ and L_0 . Let $\bar{x} = \frac{1}{N} \sum_i x^i$ and $\hat{\Sigma}_X = \frac{1}{N} \sum_i (x^i - \bar{x})(x^i - \bar{x})^\top$. In the linear-Gaussian instance used experimentally, $y = Ax + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_{\text{obs}}^2 I_k)$, which yields a Gaussian analytic posterior with covariance Σ_{post} .

3 THEORY

Scope marker. Sections 3.1–3.4 characterise the exact population minimiser over L^2 -measurable vector fields for the empirical data law, with $\mathcal{D}$ fixed; Section 3.5 quantifies one reason a finite trained network differs. Two results are the exception and change the probabilistic setting deliberately: Propositions 8 and 12 treat $\mathcal{D}$ as an i.i.d. sample and are asymptotic in N , because their content is precisely what happens as the fixed- N picture is left behind; we mark the switch where it occurs. We state the load-bearing results here and defer all proofs to Appendix A. The projection identity underlying everything is standard.

Proposition 1 (L^2 projection). *For every square-integrable v , $L_h(v) = \mathbb{E}[\text{Var}(U \mid X_t, t, \tilde{Y})] + \mathbb{E}[v - \mathbb{E}[U \mid X_t, t, \tilde{Y}]]^2$, so the a.e.-unique minimiser is $v_h^*(x, t, y) = \mathbb{E}[U \mid X_t = x, t, \tilde{Y} = y]$ with $\inf_v L_h = \mathbb{E}[\text{Var}(U \mid X_t, t, \tilde{Y})]$.*

Two facts make the conditional expectations legitimate and the flows well posed without any additional assumption (Appendix A): (i) for $t < 1$ the map $X_0 \mapsto X_t = (1-t)X_0 + tx^i$ is an invertible affine change of variables, with $X_0 = (X_t - tx^i)/(1-t)$ and conditional density $p(x \mid I = i, t) = (1-t)^{-d} \pi_0(\frac{x-tx^i}{1-t}) > 0$ for every x (Lemma 1); (ii) any field of the form $v(x, t) = \sum_i w_i(x, t)(x^i - x)/(1-t)$ with smooth weights summing to one is C^∞ hence locally Lipschitz on $\mathbb{R}^d \times [0, 1)$, so its ODE has a unique solution up to $t = 1$ and endpoint laws are the well-defined weak limits $p_1 = \lim_{t \uparrow 1} p_t$ (Lemma 2).

3.1 HARD CONDITIONING: EXACT COLLAPSE

We begin with the base case: a raw, unsmoothed label ($h = 0$) that identifies its training example uniquely. Proposition 1 already tells us the exact minimiser is a conditional expectation; the question is what that conditional expectation collapses to once the conditioning event pins down a single index i .

Proposition 2 (Exact conditional collapse; $h = 0$). *Assume $y^1, \dots, y^N$ are pairwise distinct. Then (a) $v_0^*(x, t, y^i) = (x^i - x)/(1-t)$ for $t < 1$; (b) $\inf_v L_0(v) = 0$; (c) the ODE $\dot{x}_t = (x^i - x_t)/(1-t)$ has the explicit solution $x_t = (1-t)x_0 + tx^i$, so $x_t \rightarrow x^i$ for every x_0 ; and (d) $p_1^{\text{cond}}(\cdot \mid y^i) = \delta_{x^i}$, hence $\text{Cov}[p_1^{\text{cond}}(\cdot \mid y^i)] = 0$.*

Proof. (a) Distinctness makes $\{Y = y^i\}$ identify $I = i$, so $X_1 = x^i$ is deterministic and, by Lemma 1, $X_0 = (x - tx^i)/(1-t)$ is determined by (x, t, i) . Hence $U = x^i - X_0 = (x^i - x)/(1-t)$ is $\sigma(X_t, t, Y)$ -measurable and equals its conditional expectation. (b) $U - v_0^* = 0$ a.s. (c) Multiplying by $(1-t)$ gives $\frac{d}{dt}[\frac{x_t - x^i}{1-t}] = 0$, so $\frac{x_t - x^i}{1-t} \equiv x_0 - x^i$. (d) The endpoint is x^i independently of x_0 . $\square$

If labels are not distinct, collapse is onto the label’s empirical conditional support: for $\mathcal{I}_y = \{i : y^i = y\}$, $p_1^{\text{cond}}(\cdot \mid y) = |\mathcal{I}_y|^{-1} \sum_{i \in \mathcal{I}_y} \delta_{x^i}$. Collapse is always to the empirical conditional law, which is a single atom exactly when the label is an identifier.

Proposition 3 (Unconditional endpoint is the full empirical law). *For $L_{\text{unc}}(v) = \mathbb{E}|X_1 - X_0 - v(X_t, t)|^2$ and $t < 1$, with $q_i(x, t) = \Pr(I = i \mid X_t = x, t) \propto \pi_0(\frac{x-tx^i}{1-t})$,*

$$v_{\text{unc}}^*(x, t) = \sum_i q_i(x, t) \frac{x^i - x}{1-t}, \quad p_1^{\text{unc}} = \hat{\rho}_X = \frac{1}{N} \sum_i \delta_{x^i}.$$

Moreover $\inf_v L_{\text{unc}} > 0$ whenever $x^i \neq x^j$ for some $i \neq j$.

Corollary 1 (The correct conditional/unconditional distinction). *Both exact population flows are supported on the training set: $p_1^{\text{cond}}(\cdot | y^i) = \delta_{x^i}$ and $p_1^{\text{unc}} = \frac{1}{N} \sum_i \delta_{x^i}$. The distinction is therefore not “memorises vs. does not” but single-example vs. full-empirical-measure memorisation, with observable difference in the conditional second moment:*

$$\text{Cov}[p_1^{\text{cond}}(\cdot | y^i)] = 0 \quad \text{vs.} \quad \text{Cov}[p_1^{\text{unc}}] = \hat{\Sigma}_X. \quad (2)$$

The often-cited observation $\text{tr Cov} \approx \text{tr } \hat{\Sigma}_X$ for the unconditional baseline is precisely (2), i.e. evidence of full-empirical-measure memorisation, *not* of non-memorisation.

3.2 LABEL SMOOTHING IS EXACTLY KERNEL REGRESSION

The natural fix for a label that identifies its example too precisely is to blur it: replace y^i by $\tilde{Y} = y^i + h\varepsilon$. This is not a heuristic—the population minimiser becomes exactly the velocity field of a Nadaraya–Watson kernel regression over the training atoms.

Proposition 4 (Kernel-weighted population minimiser). *For $h > 0$, $t < 1$,*

$$v_h^*(x, t, y) = \sum_i w_i^{(h)}(x, t, y) \frac{x^i - x}{1 - t}, \quad w_i^{(h)} = \frac{\pi_0\left(\frac{x - tx^i}{1 - t}\right) K_h(y - y^i)}{\sum_j \pi_0\left(\frac{x - tx^j}{1 - t}\right) K_h(y - y^j)}. \quad (3)$$

Equation (3) factorises the index posterior into a spatial source-consistency term $\pi_0\left(\frac{x - tx^i}{1 - t}\right)$ and a label kernel $K_h(y - y^i)$. Write $p_i^{(h)}(y) := K_h(y - y^i) / \sum_j K_h(y - y^j)$ for the normalised label weights. Lemma 3 (appendix) identifies v_h^* with the conditional velocity of the coupling $I \sim p^{(h)}(y)$, $X_1 = x^I$, which is what lets us read off the endpoint law:

Theorem 1 (Endpoint law under label smoothing). *For every $h > 0$ and query y ,*

$$p_h^{\text{gen}}(\cdot | y) = \sum_{i=1}^N p_i^{(h)}(y) \delta_{x^i}. \quad (4)$$

Corollary 2 (Zero bandwidth at fixed N). *If y has a unique nearest label $i^* = \arg \min_i |y - y^i|$, then $p_i^{(h)}(y) \rightarrow \mathbf{1}\{i = i^*\}$ as $h \rightarrow 0$, so $p_h^{\text{gen}}(\cdot | y) \Rightarrow \delta_{x^{i^*}}$; at $y = y^i$ this recovers Proposition 2.*

Corollary 3 (Infinite bandwidth at fixed N). *$p_i^{(h)}(y) \rightarrow 1/N$ for every i as $h \rightarrow \infty$, so $p_h^{\text{gen}}(\cdot | y) \Rightarrow p_1^{\text{unc}}$.*

So h interpolates continuously between the two hard regimes of Section 3.1: $\delta_{x^{i^*}} \rightarrow \sum_i p_i^{(h)}(y) \delta_{x^i} \rightarrow \frac{1}{N} \sum_i \delta_{x^i}$ as h grows from 0 to ∞ .

Proposition 5 (Exact finite- N moments). *$\bar{x}_h(y) = \sum_i p_i^{(h)}(y) x^i$ and $\text{Cov}_h(y) = \sum_i p_i^{(h)}(y) (x^i - \bar{x}_h(y))(x^i - \bar{x}_h(y))^\top$, both exactly computable from the training set for any h . They are the correct reference against which generated moments must be compared. $\text{tr Cov}_h(y)$ need not be monotone in h ; monotonicity is not implied and must be reported empirically.*

Proposition 6 (Label smoothing never leaves the training set). *For every $h \in [0, \infty)$, $\text{supp}(p_h^{\text{gen}}(\cdot | y)) \subseteq \{x^1, \dots, x^N\}$. Consequently, if the true posterior $p(\cdot | y)$ is absolutely continuous,*

$$W_2^2(p_h^{\text{gen}}(\cdot | y), p(\cdot | y)) \geq \int \text{dist}(x, \{x^i\})^2 p(x | y) dx > 0, \quad (5)$$

and this lower bound does not depend on h .

This is the central caveat on any “variance is restored at $h \approx 0.1$ ” claim: label smoothing *redistributes weight over training atoms*; it never generates new samples, so matching tr Cov_h matches a second moment, not the posterior. A genuine remedy needs $h \rightarrow 0$ and $N \rightarrow \infty$ jointly—the Nadaraya–Watson consistency regime of Proposition 12 (appendix)—not h alone.

3.3 ESCAPING THE OBSTRUCTION: SMOOTHING THE ENDPOINT

Proposition 6 rules out any remedy leaving the endpoint law on the training atoms—label smoothing, and trivially the interpolant noise of Section 3.4 since $\gamma(1) = 0$ —and says what a remedy must do: move the support. Of the three places to inject noise (label, path, endpoint), only the third does.

Proposition 7 (Endpoint smoothing). *Replace the endpoint by $X_1 = x^I + \rho \xi$ with $\xi \sim \mathcal{N}(0, I_d)$ independent of (I, X_0, ε) and redrawn every step, keeping label bandwidth $h \geq 0$. Write $s_t^2 = (1-t)^2 + t^2 \rho^2$. Then v_h^* is again a weighted average of per-atom fields, now with spatial weights $\varphi_{s_t}(x - tx^i)$ in place of $\pi_0(\frac{x-tx^i}{1-t})$ (explicitly, (19) in the appendix), and the endpoint law is*

$$p_1(\cdot | y) = \sum_i p_i^{(h)}(y) \mathcal{N}(x^i, \rho^2 I_d), \quad \bar{x} = \bar{x}_h(y), \quad \text{Cov} = \text{Cov}_h(y) + \rho^2 I_d. \quad (6)$$

For every $\rho > 0$ this law is absolutely continuous, so (5) does not apply to it; moreover $s_1 = \rho > 0$, so v_h^* extends continuously to $t = 1$ and the $1/(1-t)$ blow-up of $(\star)$ —the source of both Lemma 5 and the fixed-learning-rate divergence of Section 4.1—disappears.

Two caveats, both quantified in Section 4.2: removing a lower bound is not reducing the distance—(6) is a kernel density estimate, paying $+\rho^2 I_d$ of variance for its continuity—and the gain is bounded by the obstruction removed, the floor scaling like $N^{-2/d}$. What is unconditional is the second knob: h sets $\bar{x}_h$ and Cov_h , ρ shifts the covariance by $\rho^2 I_d$, so together they match the posterior’s first and second moments with a continuous law, where h alone hits at most one while staying atomic.

The next two results are the one place we leave the fixed- $\mathcal{D}$ setting: they ask what Cov_h and $\bar{x}_h$ converge to when the training set is itself an i.i.d. sample and $N \rightarrow \infty$, which is the only regime in which label smoothing is a genuine remedy rather than a reweighting.

Proposition 8 (Bandwidth expansion of the conditional covariance; random design). *Let (x^j, y^j) be i.i.d. with marginal label density ρ_Y , $\Sigma(y) = \text{Cov}(X | Y = y)$ and $J(y) = \partial \mu / \partial y$. Fix a query point y in the interior of $\text{supp}(\rho_Y)$ at which ρ_Y is positive and C^2 , $\Sigma(\cdot)$ is C^1 , and $\mu(\cdot)$ is C^2 (all standard nonparametric-regression regularity, needed only in a neighbourhood of this fixed y). Then, pointwise at this y , as $Nh^k \rightarrow \infty, h \rightarrow 0$,*

$$\text{Cov}_h(y) = \Sigma(y) + h^2 J(y) J(y)^\top + O(h^4) + O_P((Nh^k)^{-1/2}), \quad (7)$$

so $\text{tr Cov}_h(y) = \text{tr } \Sigma(y) + h^2 \|J(y)\|_F^2 + \dots$. The expansion is not claimed uniform over y ; near the boundary of $\text{supp}(\rho_Y)$ the kernel loses mass asymmetrically and the bias term changes order (the standard nonparametric boundary-bias phenomenon), which (7) does not cover.

The between-group inflation $h^2 \|J\|_F^2$ is positive and, unlike the bias of the conditional *mean*, is not annihilated by kernel symmetry; it produces the right-hand (over-smoothing) branch of the empirical U-curve. In the linear-Gaussian model $\mu(y)$ is exactly linear, so $J = \Sigma_{\text{post}} A^\top / \sigma_{\text{obs}}^2$ is known in closed form and the $O(h^4)$ term vanishes: (7) is then a parameter-free prediction. Proposition 12 (appendix) gives the companion Nadaraya–Watson rates for $\bar{x}_h(y)$, with $h^* \asymp N^{-1/(k+4)}$, and records why queries at $y = y^i$ are *design points*: there x^i carries the maximal weight $K_h(0)$, so this $h \rightarrow 0$ limit is a genuinely different one from Corollary 2’s fixed- N collapse—same symbol, opposite conclusions.

3.4 STOCHASTIC INTERPOLANT NOISE DOES NOT HELP

An equally natural place to inject randomness is the interpolant path itself, as in stochastic-interpolant constructions. This remedy is qualitatively different: it never touches the population endpoint, at any noise level. Replace the interpolant by $X_t = (1-t)X_0 + tX_1 + \gamma(t)Z$ with $Z \sim \mathcal{N}(0, I_d)$ independent and $\gamma(t) = \sigma \sqrt{t(1-t)}$, so $\gamma(0) = \gamma(1) = 0$. A single Z defines the whole trajectory, and the correct pathwise-derivative target is

$$\dot{X}_t = X_1 - X_0 + \dot{\gamma}(t)Z, \quad \dot{\gamma}(t) = \frac{\sigma(1-2t)}{2\sqrt{t(1-t)}}. \quad (8)$$

Regressing on $X_1 - X_0$ alone is a *different* objective and does not yield the marginal velocity of the noisy path. Let $s_t^2 := (1-t)^2 + \gamma(t)^2 = (1-t)[(1-t) + \sigma^2 t]$, so $s_0 = 1, s_1 = 0$.

Proposition 9 (Closed form and endpoint invariance in σ). *Let $h = 0$ with distinct labels and $v_\sigma^* = \mathbb{E}[\dot{X}_t \mid X_t, t, Y]$ for target (8). Then for $t < 1$: (a) $v_\sigma^*(x, t, y^i) = x^i + c(t)(x - tx^i)$ with $c(t) = \frac{-(1-t) + \frac{\sigma^2}{2}(1-2t)}{s_t^2} = \frac{1}{2} \frac{d}{dt} \log s_t^2$; (b) the flow is $x_t = t x^i + s_t x_0$; and (c) since $s_1 = 0$,*

$$p_1^{\text{cond}}(\cdot \mid y^i) = \delta_{x^i} \quad \text{for every } \sigma \geq 0. \quad (9)$$

Corollary 5 (appendix) records what interpolant noise *does* change: it raises the irreducible error ($\inf_v L_\sigma > 0$ for $\sigma > 0$, whereas $\inf_v L_0 = 0$) and halves the Lipschitz blow-up rate. Neither touches p_1 .

Proposition 10 (The two mechanisms act on different factors). *In the factorisation (3), label smoothing modifies the label factor $K_h(y - y^i)$ and therefore changes p_1 (Theorem 1); interpolant noise rescales the spatial factor $\mathcal{N}(tx^i, s_t^2 I_d)$ isotropically and identically for every i , leaving the endpoint law unchanged (Proposition 9). The two remedies are therefore not interchangeable.*

This cleanly separates the present conditional mechanism from the unconditional “resample- x_0 ” remedy of the gradient-variance line (Reu et al., 2025), which operates through attainability rather than through the population optimum.

3.5 FINITE CAPACITY: A GENUINE REPRESENTATION FLOOR

The results so far describe the population minimiser over *all* square-integrable fields; a trained network is restricted to a finite-capacity class. Here is one concrete way that restriction alone forces a positive loss floor, independent of any optimisation difficulty.

Proposition 11 (Lipschitz lower bound). *Fix $i, t < 1$, $f_i(x, t) = (x^i - x)/(1 - t)$, and suppose $v(\cdot, t, y^i)$ is L -Lipschitz in x . Under $X_t \mid I = i \sim \mathcal{N}(tx^i, (1-t)^2 I_d)$, $\mathbb{E}[|v(X_t, t, y^i) - f_i(X_t, t)|^2 \mid I = i] \geq d[1 - L(1 - t)]_+^2$.*

Corollary 4 (Loss floor under a uniform Lipschitz constraint). *For $\mathcal{F}_L = \{v : \text{Lip}_x v(\cdot, t, y^i) \leq L \forall t, i\}$ with $L \geq 1$ and $t \sim \text{Unif}(0, 1)$,*

$$\inf_{v \in \mathcal{F}_L} L_0(v) \geq d \int_{1-1/L}^1 [1 - L(1 - t)]^2 dt = \frac{d}{3L} > 0. \quad (10)$$

Two warnings temper (10). First, the bound is *conditional* on the constraint: an MLP with unbounded weights lies in no fixed $\mathcal{F}_L$, and by universal approximation $\inf_\theta L(v_\theta)$ may be 0; what we prove is that any architecture with $\text{Lip}_x v_\theta \leq L$ cannot beat $d/(3L)$. Second, it does not bind in our experiments: with $d = 2$ and a trained Lip_x of order 10^1 – 10^2 , the floor is of order 10^{-2} , one to two orders of magnitude below the observed plateau 0.48. Corollary 4 is therefore best used as a *measurement*—estimate L , compare $d/(3L)$ to the plateau, report which gap dominates (Appendix C.9)—and that measurement rules the representation explanation *out* here.

3.6 FINITE-SAMPLE MEMORISATION (SUMMARY)

Appendix B adapts two representability results of the unconditional gradient-variance literature (Reu et al., 2025) to the present notation—a fixed finite *unconditional* batch always admits an exact zero-loss memorising field despite Proposition 3’s positive population loss (Proposition 13), and resampling x_0 every epoch is exactly what defeats that construction (Remark 1)—and adds a new, problem-specific result: no finite-depth, finite-width network with globally Lipschitz activations can equal the collapsed field’s $1/(1 - t)$ blow-up *pointwise* on all of $[0, 1)$ (Lemma 5). We keep that pointwise claim separate from a loss floor: the only class for which we prove a strictly positive floor is the Lipschitz-capped $\mathcal{F}_L$ of Corollary 4. The contrast with Proposition 2 is the point—conditional collapse needs no genericity argument and no fixed-batch caveat, because y^i pins down i for every x_0 , not merely with probability one on a single draw.

3.7 SCOPE: WHAT IS AND IS NOT PREDICTED

The population theory proves exact consequences for the empirical minimiser (Propositions 2–8, Corollaries 2–3, Proposition 11, Corollary 4); Appendix B adds exact statements about a *randomly*

drawn finite unconditional batch. It does *not* determine: the number of SGD steps to approach the optimum; the dependence of *observed* collapse on N at fixed architecture (the population optimum collapses for every finite N , so a non-collapsing $N=5000$ run probes representation/optimisation, it does not contradict Proposition 2); the dependence on σ_{obs} (the theory implies no monotonicity, so a flat sweep is *consistent with*, not a falsification of, the theory); the monotonicity of tr Cov_h in h ; or finite-sample error bars. We therefore never call a quantity in that second list a “theoretical prediction” that “matched” or “failed”; it is reported as an empirical finding, marked “(empirical).”

4 EXPERIMENTS

Setup. Unless noted, EXP-1 uses the linear-Gaussian problem $d=2$, $k=1$, $N=200$, $\sigma_{\text{obs}}=0.1$, prior and source $\mathcal{N}(0, I)$, deterministic linear interpolant ($\sigma=0$); a 4-layer width-128 MLP with SiLU and a sinusoidal time embedding (dim 64); Adam at $\text{lr}=10^{-3}$, batch 256; RK4 integration with 100 steps stopping at $t=1-10^{-3}$; evaluation over $M=1000$ samples at 20 training and 8 held-out conditions. We do *not* early-stop: overtraining is the object of study. Sanity checks pass first (the analytic posterior satisfies the normal equations to 1.8×10^{-15} ; closed-form field integration maps every $x_0 \rightarrow x^i$; the unconditional model does not leak y). Sweeps use five seeds per point. The seed controls *both* the problem instance (the operator A , hence Σ_{post}) *and* all training randomness, so every error bar below mixes the two; we flag this conflation rather than disentangle it.

4.1 EXP-1: EXACT-THEORY QUANTITIES (P1–P4)

Table 1 reports the four exact-theory observables (curves in Figures 8 and 9, appendix). As training proceeds past the good-calibration phase, the conditional generated variance collapses from $\approx \Sigma_{\text{post}}$ toward 0 (P1), the velocity field approaches the collapsed closed form $(x^i - x)/(1 - t)$ (P2), the generated mean moves from μ_{post} toward the training point x^i (P3), and the unconditional baseline retains data-scale variance throughout (P4).

quantity	iteration				
	10^2	10^4	10^5	2×10^5	1×10^6
conditional tr Cov	1.08 ± 0.02	1.03 ± 0.11	0.58 ± 0.18	0.396 ± 0.127	0.120 ± 0.054
velocity rel. error vs. $(\star)$	0.78 ± 0.03	0.75 ± 0.07	0.50 ± 0.04	0.40 ± 0.05	0.168 ± 0.052
$\ \text{mean} - x^i\ $	0.90 ± 0.16	0.87 ± 0.20	0.58 ± 0.13	0.38 ± 0.10	0.137 ± 0.065
$\ \text{mean} - \mu_{\text{post}}\ $	0.15 ± 0.06	0.25 ± 0.07	0.52 ± 0.10	0.66 ± 0.18	0.816 ± 0.172
unconditional tr Cov	1.85 ± 0.55	1.93 ± 0.16	1.91 ± 0.17	1.926 ± 0.171	—

Table 1: EXP-1, mean $\pm$ std over 5 seeds; the 10^6 column is the cosine-schedule run (no unconditional counterpart), whose between-seed spread makes it evidence of *stability* rather than a precise depth estimate. Same-seed reference values: $\text{tr } \Sigma_{\text{post}} = 1.018 \pm 0.016$, $\text{tr } \hat{\Sigma}_X = 1.931 \pm 0.162$ —the unconditional row matches the latter almost exactly, which is (2) measured directly.

The variance tracks the true posterior up to $\sim 10^4$ iterations—the model is a correct Bayesian sampler in this phase, which is exactly why early stopping works (Dasgupta et al., 2026)—and then collapses, stalling near 0.40 at 2×10^5 . This residual is an *optimisation* gap, not a failure of Proposition 2: all four observables keep descending with the loss. At a *fixed* learning rate, however, training eventually diverges near 10^6 iterations, since $(\star)$ steepens as $t \rightarrow 1$ and a constant step becomes too large once the loss is small. A cosine schedule ($\text{lr} : 10^{-3} \rightarrow 10^{-6}$) removes that divergence on all 5 seeds. The cleanest evidence of deeper collapse is same-seed: at the identical 7×10^5 checkpoint seed 0 reaches $\text{tr Cov} = 0.128$ with the schedule versus 0.162 without, continuing to 0.115 by 10^6 . The 5-seed mean 0.120 ± 0.054 shows mainly that no seed diverges, *not* a tight improvement on 0.16: seeds at 10^6 range 0.035–0.186, a spread inflated by the seed-conflated problem instance, and only four of five improve from 7×10^5 to 10^6 . Figure 10 (appendix) shows the endpoint in sample space, with per-condition tr Cov spanning 5.8×10^{-5} to 8.4×10^{-2} —collapse in every condition, though not to equal depth.

A literature-standard memorisation ratio confirms Corollary 1. The diagnostics above are aggregate (a covariance trace, or distances of the sample *mean*). As a per-sample complement we re-

port the *memorisation ratio* of Yoon et al. (2023)¹: $\hat{x}$ is *memorised* if $\|\hat{x} - x^{(1)}\|^2 \leq c\|\hat{x} - x^{(2)}\|^2$ for $c = 1/9$, with $x^{(1)}, x^{(2)}$ the two nearest training points. Over 5 seeds both flows rise from a common ≈ 0.11 baseline, the conditional reaching 0.650 ± 0.033 at 2×10^5 versus 0.389 ± 0.032 unconditionally (Figure 12)—confirmation that *both* memorise (Corollary 1), conditionally more strongly. The two are not comparable in magnitude—the unconditional target is spread over $N=200$ atoms, so the $c=1/9$ test is mechanically harder there—so we read this directionally only. The scheduled run reaches 0.955 ± 0.020 by 10^6 : 95% of *individual* samples are memorised, not merely the sample mean.

Held-out labels: nearest-atom collapse, not generalisation. Corollary 2 predicts what the diagnostics above do not test: at a query y that is *not* a training label the $h \rightarrow 0$ endpoint is $\delta_{x^{i^*}}$ for the nearest label i^* , so held-out conditions should collapse too—onto a memorised atom, not onto the correct posterior. Over 8 held-out y per run this is what happens: held-out tr Cov tracks the training value (medians 0.43 vs. 0.42 at 2×10^5) while $\|\text{mean} - \mu_{\text{post}}(y)\|$ grows from 0.12 to 0.87 (median): overtraining moves the sampler away from the Bayes answer at *new* labels too. We quote medians because in one of five seeds the held-out ODE *diverges*, inflating the mean to 6×10^4 (Appendix C.6).

Adversarial pairing: collapse does not require a real posterior. Proposition 2 never uses A or σ_{obs} —only that the labels are pairwise distinct—so collapse should be unchanged if Y is randomly permuted relative to X before training. It is: tr Cov falls to 0.625 ± 0.108 at 2×10^5 (vs. 0.396 ± 0.127 under the real pairing, an optimisation-difficulty gap at fixed budget), while the distance to y^i ’s *true* posterior mean *grows*, $0.84 \pm 0.13 \rightarrow 1.437 \pm 0.176$, crossing above $\|\text{mean} - x^i\|$ between 10^4 and 3×10^4 iterations. The generated law tracks the arbitrary label-to-example assignment, not any statistical relation between x and y (Appendix C.2).

4.2 LABEL SMOOTHING TRACKS THE KERNEL REFERENCE (P7)

This is our strongest empirical confirmation of the theory. The correct reference is Cov_h (Theorem 1, Proposition 5), *not* Σ_{post} . Evaluating trained checkpoints against the exact kernel moments over 20 conditions ($M=1000$, 5 seeds), the generated variance tracks Cov_h at every bandwidth: ratio-to-kernel = 1.002 ± 0.050 , 0.994 ± 0.039 and 1.011 ± 0.022 at $h = 0.05, 0.1, 0.5$ (Figure 1; Table 10, appendix, gives all entries). This is a verification of the *endpoint law*, stronger than a velocity-field check. At $h=0.01$ the ratio is high and unstable (1.308 ± 0.376) because $n_{\text{eff}} \approx 3.5$ leaves Cov_h near-degenerate for many conditions; the ratio of mean traces, $0.673/0.583 = 1.15$, is closer to 1.

A natural but mistaken reading is that $h \approx 0.1$ “perfectly restores” the posterior variance: tr Cov merely crosses $\Sigma_{\text{post}} \approx 1.02$ near $h=0.1$ because for this problem $\Sigma_{\text{post}} \approx \text{Cov}_h|_{h=0.1} \approx 1.00$ —a coincidence. The correct target is Cov_h , which the model attains at *every* h . The field itself agrees: v_θ matches the kernel minimiser (3) 2–4 $\times$ better than the single-atom field ($\star$) at every h , and the gap widens with h (Table 11 and Figure 15, appendix).

Atomicity caveat (empirical confirmation of Proposition 6). Even at the best-matching h the generated law remains atomic: $n_{\text{eff}} \approx 26/200$ at $h=0.1$ and $\approx 102/200$ at $h=0.5$, and the MMD to the analytic posterior falls with h ($0.256 \rightarrow 0.012$) but reaches zero at no h (Appendix C.9). Matching variance is not recovering the posterior.

How large is the atomicity floor, and does endpoint smoothing help? Unlike the MMD, the floor (5) is directly computable: at the EXP-1 configuration $F = 0.0353$, i.e. $W_2 \geq 0.19$ but only 3.5% of $\text{tr } \Sigma_{\text{post}}$, and the *population-optimal* atomic law already reaches $\text{MMD} = 0.0089$ —below the trained model’s 0.024 . Atomicity, though exact, is therefore *not* what the MMD curves measure; we correct that misreading in Appendix C.9. The reason is instrumental: an RBF MMD at the median-heuristic bandwidth cannot resolve inter-atom spacing. Scored in an entropic OT divergence, which can, endpoint smoothing (Proposition 7) cuts the population distance to the posterior $2.1\times$ at $h=0.1$ ($12.0 \rightarrow 5.7$ at $\rho=0.1$), with the expected U-shape in ρ , while MMD moves by at most $1.2\times$. A *trained model attains this*: retraining EXP-1 at $N=50$ (where the $d=2$ floor is largest, $F/\text{tr } \Sigma_{\text{post}} = 0.19$) with $h=0.1$ and $\rho \in \{0, 0.3\}$, the generated tr Cov matches (6)’s

¹As used, e.g., by Buchanan et al. (2025).

$\text{tr Cov}_h + d\rho^2$ to within 3% in both arms and the mean is unchanged, while the OT distance falls $2.5\times$ ($79.2 \rightarrow 31.5$, and $2.5\times$ in each seed) against only $1.5\times$ in MMD (Appendix C.5). Table 3 sweeps (d, N) and adds the caveat that matters: the gain grows with the floor at fixed d but *not* across d —at $d=10$ the floor is largest and the gain among the smallest, since an isotropic ρ cannot fill a high-dimensional gap without paying $d\rho^2$. Endpoint smoothing is defeated by dimension, as kernel density estimation is.

Interpolant noise does not help (Proposition 9). With the *corrected* target (8) (verified by four closed-form checks) the generated variance is unchanged from that obtained with the mis-specified $X_1 - X_0$ target (0.367 ± 0.146 vs. 0.372 ± 0.140 at $\sigma=0.1$; Table 12, appendix) and stays far below the posterior—exactly as (9) predicts, the endpoint law being δ_{x^i} for every σ whether or not the target is correct.

4.3 EXP-2 AND EXP-3: SELECTIVE MEMORISATION BEYOND THE GAUSSIAN CASE

EXP-2 (Gaussian mixture). A 4-mode GMM prior with a rank-deflating linear observation gives a genuinely *bimodal* analytic posterior. Up to $\sim 10^4$ iterations mode coverage is 1.0 and the MMD near zero—a calibrated sampler—after which coverage falls to 0.72 and the MMD grows to ~ 0.25 (Fig. 16), while the share of late samples nearest the x^i -containing mode rises from ≈ 0.5 to 0.999/0.766/0.990/1.000 (Fig. 17): selective memorisation.

EXP-3 (image inpainting). Masking the lower half of 32×32 MNIST ($N=500$) with a small U-Net, over 3 seeds the inpainted-region pixel variance drops $\sim 100\times$ by iteration 15000 ($0.137 \pm 0.021 \rightarrow (1.17 \pm 0.12) \times 10^{-3}$) and the nearest-training-image distance falls to $(0.77 \pm 0.26) \times 10^{-3}$: completions diverse at iteration 200 become identical by 15000 and coincide with that image (Fig. 19). CIFAR-10 (one seed) gives the same collapse, $0.142 \rightarrow 0.0087$ (Fig. 20).

5 LIMITATIONS

The optimisation gap is bounded away from one explanation, not explained. A finite network reaches the population minimiser only up to an optimisation gap, which is why hard-conditioning collapse is *optimisation-paced* and only partial within a finite budget (for $h > 0$ the model does reach Cov_h). Corollary 4’s Lipschitz floor was our candidate explanation for the $h=0$ plateau, and Appendix C.9 shows it does *not* bind: it is $\sim 50\times$ below the plateau, which is moreover strongly schedule-sensitive (Table 1). We have thus ruled *this* mechanism out without supplying a positive quantitative theory of the gap itself.

Small scale, and a remedy of limited reach. EXP-1/EXP-2 are $d \leq 10$ synthetic problems and EXP-3 uses 32×32 images, $N=500$ and one small U-Net (3 MNIST seeds, 1 CIFAR-10 seed); the mechanism is parameterisation-independent by construction, but persistence *at scale* is a separate claim we do not establish, and the width/depth/optimiser sweep (Appendix C) covers only tr Cov at one budget. Endpoint smoothing (Proposition 7) removes the atomicity obstruction by construction and is confirmed on a trained model at $N=50$, but its benefit is bounded by that obstruction’s size—small in exactly the regime we study most—it buys continuity at the price of $+\rho^2 I_d$ of variance, and it degrades with dimension (Table 3); we have not tested it at scale, on EXP-2/EXP-3, or with an anisotropic ρ . Finally, P5/P6 are empirical findings the theory does not determine, the seed controls both problem instance and training run, and the checkpoint-based tables require retraining the $p7y$ runs.

6 CONCLUSION

Conditional collapse in flow matching is the zero-bandwidth limit of a kernel-weighted empirical conditional flow, and label smoothing makes that flow exact kernel regression over the training atoms. Neither label nor interpolant noise leaves the atomic support; endpoint smoothing does, but only as far as the floor it removes is large.

REPRODUCIBILITY STATEMENT

All numbers reported are produced by code included in the supplementary material (anonymized for review). Theory checks: `test_kernel_theory.py` (reference table to $< 0.05\%$), `analyze_p7_kernel.py`, `verify_kernel_theory.py`, `verify_prop17.py`, `verify_target_noise.py`, `atomicity_scaling.py`. Experiments: `run_exp1.sh`, `run_sweeps.py` (`--gpus N`), `train_exp2`, `train_exp3`, each followed by the matching analysis script. Proofs of all load-bearing results are given in full in Section 3 and Appendix A; every stated assumption (distinct labels, smoothness of the weight functions, the linear-Gaussian instance) is made explicit at the point it is used.

ETHICS STATEMENT

This work trains small models on synthetic linear-Gaussian and Gaussian-mixture data and on the public MNIST dataset. No human subjects, personally identifiable information, or other sensitive data are used or generated. The paper’s contribution is a precise theoretical and empirical account of an existing memorisation phenomenon in conditional generative models, which bears directly on the data-privacy and copyright concerns around deployed diffusion/flow models raised in Section 1; by characterising the mechanism exactly, we aim to help downstream work design principled mitigations rather than to facilitate memorisation. We are not aware of any other ethical concerns raised by this submission.

AI USE STATEMENT

We used a generative-AI coding and writing assistant (Claude, Anthropic) throughout this project’s development: to draft and revise manuscript text, including the conversion of the manuscript to the ICLR format and multiple rounds of copy-editing; to expand several of the proofs in Section 3 and Appendix A from a theory memorandum into the propositions and proofs given here; to write the training, evaluation, and analysis code in the released repository; and to run and analyse several of the reported experiments, including the adversarial-pairing ablation (Section C.3) and the memorisation-ratio diagnostic (Section 4.1). We did not use generative AI to select, fabricate, or embellish experimental results: every number reported is produced by the released code from actual training runs. All AI-drafted proofs were checked line-by-line against the underlying mathematics by the authors, and all AI-assisted code and analysis outputs were independently verified by re-running the experiments and cross-checking the reported statistics against the raw output files. We take responsibility for the final content of this work, including all text, claims, proofs, and code.

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A DEFERRED PROOFS

Lemma 1 (Change of variables). *Fix i and $t < 1$. The map $X_0 \mapsto X_t = (1-t)X_0 + tx^i$ is invertible affine with $X_0 = (X_t - tx^i)/(1-t)$, so $\sigma(X_t | I = i) = \sigma(X_0 | I = i)$ and the conditional density is $p(x | I = i, t) = (1-t)^{-d} \pi_0\left(\frac{x-tx^i}{1-t}\right)$. The Jacobian $(1-t)^{-d}$ is i -independent and cancels in every posterior over I . Because π_0 is Gaussian, $p(x | I = i, t) > 0$ for every x and $t < 1$.*

Lemma 2 (Well-posedness). *Let $w_1, \dots, w_N : \mathbb{R}^d \times [0, 1] \rightarrow [0, 1]$ be C^∞ with $\sum_i w_i \equiv 1$ and $v(x, t) = \sum_i w_i(x, t)(x^i - x)/(1-t)$. Then v is C^∞ hence locally Lipschitz on $\mathbb{R}^d \times [0, 1]$, the ODE $\dot{x}_t = v(x_t, t)$ has a unique solution on $[0, 1]$ for every initial condition, and with $M = \max_i |x^i|$, $|x_t| + M \leq (|x_0| + M)/(1-t)$. Consequently the continuity equation has a unique measure solution and endpoint laws are the weak limits $p_1 = \lim_{t \uparrow 1} p_t$.*

Both v_{unc}^* and v_h^* have this form with smooth weights, so Lemma 2 applies and no “assume the flow is well defined” hypothesis is needed. We now give the remaining proofs in full.

Proof of Proposition 3. Equation (2)’s $q_i(x, t)$ is Bayes’ rule applied to Lemma 1’s density $p(x | I = i, t) = (1 - t)^{-d} \pi_0(\frac{x - tx^i}{1 - t})$ under the uniform prior on I : the Jacobian $(1 - t)^{-d}$ and the prior $1/N$ do not depend on i and cancel, leaving $q_i(x, t) \propto \pi_0(\frac{x - tx^i}{1 - t})$. Given $I = i$, Proposition 2(a)’s argument gives $U = (x^i - x)/(1 - t)$; averaging over the posterior $q_i(x, t)$ and substituting into Proposition 1 gives v_{unc}^* . For the endpoint law: by construction $v_{\text{unc}}^*(x, t) = \mathbb{E}[\dot{X}_t | X_t = x, t]$, which is exactly the velocity field appearing in the continuity equation for the marginal law of the interpolation. By Lemma 2 the flow is well posed on $[0, 1)$ and transports $p_0 = \pi_0$ to $p_t = \text{Law}(X_t)$ for every $t < 1$; since $X_t \rightarrow X_1 \sim \hat{\rho}_X$ almost surely, $p_t \Rightarrow \hat{\rho}_X$, giving $p_1^{\text{unc}} = \hat{\rho}_X$.

For $\inf_v L_{\text{unc}} > 0$: fix $t < 1$ and x . By the remark after Lemma 1, π_0 is everywhere positive, so $q_i(x, t) > 0$ for every i . Conditionally on $(X_t = x, t, I = i)$ the target is the deterministic vector $u_i = (x^i - x)/(1 - t)$, and $u_i \neq u_j$ whenever $x^i \neq x^j$. Hence the discrete law $\{q_i(x, t)\}_i$ places positive mass on at least two distinct values $u_i \neq u_j$ whenever some $x^i \neq x^j$, so $\text{Var}(U | X_t = x, t) = \sum_i q_i(x, t) |u_i - \bar{u}(x, t)|^2 > 0$ at every (x, t) with $t < 1$, where $\bar{u}(x, t) = \sum_i q_i(x, t) u_i$. Integrating over (x, t) and applying Proposition 1, $\inf_v L_{\text{unc}}(v) = \mathbb{E}[\text{Var}(U | X_t, t)] > 0$. $\square$

Proof of Proposition 6. The support claim is Theorem 1. For any coupling (X, X') of p_h^{gen} and $p(\cdot | y)$, $X \in \{x^i\}$ a.s. so $|X - X'| \geq \text{dist}(X', \{x^i\})$ pointwise; take expectations and infimise. The right side is the same for every h and is strictly positive because a finite set is $p(\cdot | y)$ -null. $\square$

Proof of Corollary 3. $K_h(y - y^i) = (2\pi h^2)^{-k/2} [1 + O(h^{-2})]$ uniformly in i as $h \rightarrow \infty$, so the normalised weights $p_i^{(h)}(y) \rightarrow 1/N$; apply Theorem 1. $\square$

Proof of Corollary 2. For $i \neq i^*$, $K_h(y - y^i)/K_h(y - y^{i^*}) = \exp[-(|y - y^i|^2 - |y - y^{i^*}|^2)/2h^2] \rightarrow 0$ as $h \rightarrow 0$, since $|y - y^i| > |y - y^{i^*}|$; normalising the weights $K_h(y - y^j)$ gives $p^{(h)}(y) \rightarrow \mathbf{1}\{i = i^*\}$, and Theorem 1 gives the claim. $\square$

Proof of Theorem 1. By Lemma 3, $v_h^*(\cdot, \cdot, y)$ is exactly the conditional velocity field of the mixture coupling $I \sim p^{(h)}(y)$, $X_0 \sim \pi_0$, $X_1 = x^I$, driven by the same linear interpolant. This field has the form $v(x, t) = \sum_i w_i(x, t)(x^i - x)/(1 - t)$ of Lemma 2, with smooth weights $w_i = w_i^{(h)}(x, t, y)$ summing to one (a ratio of smooth, strictly positive functions of (x, t) , by Lemma 1’s positivity of π_0), so Lemma 2 applies: the ODE $\dot{x}_t = v_h^*(x_t, t, y)$ is well posed on $[0, 1)$ for every initial condition, and by the superposition principle its flow $(\Phi_t)_{t < 1}$ transports $\pi_0 = \text{Law}(X_0)$ to $p_t := \text{Law}(X_t)$ for every $t < 1$, where X_t is the coupling’s own interpolant—because v_h^* is by construction the coupling’s conditional velocity, $(\Phi_t)_{\#} \pi_0$ and $\text{Law}(X_t)$ solve the same continuity equation with the same initial condition, hence coincide by uniqueness. It remains to identify the limiting law of X_t as $t \uparrow 1$: since $X_t = (1 - t)X_0 + tx^I$ with $X_0 \sim \pi_0$ independent of I , $X_t \rightarrow x^I$ almost surely as $t \uparrow 1$ for every realisation of I ; hence $\text{Law}(X_t) \Rightarrow \text{Law}(x^I) = \sum_i p_i^{(h)}(y) \delta_{x^i}$ weakly, which is $p_1 = \lim_{t \uparrow 1} p_t$ by Lemma 2. This is (4). $\square$

Lemma 3 (Equivalent mixture coupling). *Fix y and consider the coupling $I \sim p^{(h)}(y)$, $X_0 \sim \pi_0$ (independent of I), $X_1 := x^I$, with the same linear interpolant $X_t = (1 - t)X_0 + tX_1$. Its conditional velocity field $\mathbb{E}[X_1 - X_0 | X_t = x, t]$ equals $v_h^*(x, t, y)$ of (3) for every $x, t < 1$.*

Proof of Lemma 3. Under this coupling, $\Pr(I = i | X_t = x, t) \propto \Pr(I = i)p(x | I = i, t) = p_i^{(h)}(y) \cdot (1 - t)^{-d} \pi_0(\frac{x - tx^i}{1 - t})$ by the same Bayes’-rule computation as above (now with prior $p_i^{(h)}(y)$ in place of $1/N$), and $p_i^{(h)}(y) \propto K_h(y - y^i)$ by definition, so $\Pr(I = i | X_t = x, t) \propto \pi_0(\frac{x - tx^i}{1 - t}) K_h(y - y^i)$ —the same proportionality (in i , at fixed x, t, y) as the numerator of $w_i^{(h)}$ in (3); normalising over i recovers $w_i^{(h)}(x, t, y)$ exactly. As in the proof above, given $I = i$ the target is $(x^i - x)/(1 - t)$, so $\mathbb{E}[X_1 - X_0 | X_t = x, t] = \sum_i w_i^{(h)}(x, t, y)(x^i - x)/(1 - t) = v_h^*(x, t, y)$. $\square$

Proof of Proposition 4. Fix (x, t, y) with $t < 1$. By Bayes' rule applied to the discrete latent index I ,

$$\Pr(I = i \mid X_t = x, t, \tilde{Y} = y) = \frac{\Pr(I = i) p(x, y \mid I = i, t)}{\sum_j \Pr(I = j) p(x, y \mid I = j, t)}. \quad (11)$$

Given $I = i$, $X_t = (1 - t)X_0 + tx^i$ is a deterministic affine function of X_0 alone and $\tilde{Y} = y^i + h\varepsilon$ is a deterministic affine function of ε alone, with $X_0 \perp \varepsilon$; hence $X_t \perp \tilde{Y} \mid I = i$, with marginal conditional densities $p(x \mid I = i, t) = (1 - t)^{-d} \pi_0\left(\frac{x - tx^i}{1 - t}\right)$ (Lemma 1) and $p(y \mid I = i) = K_h(y - y^i)$ (the density of $y^i + h\varepsilon$). Independence gives the joint conditional density $p(x, y \mid I = i, t) = (1 - t)^{-d} \pi_0\left(\frac{x - tx^i}{1 - t}\right) K_h(y - y^i)$; with the uniform prior $\Pr(I = i) = 1/N$, (11) becomes

$$\begin{aligned} \Pr(I = i \mid X_t = x, t, \tilde{Y} = y) &= \frac{\frac{1}{N} (1 - t)^{-d} \pi_0\left(\frac{x - tx^i}{1 - t}\right) K_h(y - y^i)}{\sum_j \frac{1}{N} (1 - t)^{-d} \pi_0\left(\frac{x - tx^j}{1 - t}\right) K_h(y - y^j)} \\ &= \frac{\pi_0\left(\frac{x - tx^i}{1 - t}\right) K_h(y - y^i)}{\sum_j \pi_0\left(\frac{x - tx^j}{1 - t}\right) K_h(y - y^j)} = w_i^{(h)}(x, t, y), \end{aligned}$$

the i -independent factor $\frac{1}{N} (1 - t)^{-d}$ cancelling between the numerator and every term of the denominator (the kernel normalisation constant inside K_h cancels identically for the same reason). Given $I = i$, one further conditions on $X_t = x$: by Lemma 1 this determines $X_0 = (x - tx^i)/(1 - t)$, so $U = X_1 - X_0 = x^i - X_0 = (x^i - x)/(1 - t)$, exactly as in Proposition 2(a). By Proposition 1, the population minimiser is $v_h^*(x, t, y) = \mathbb{E}[U \mid X_t = x, t, \tilde{Y} = y]$, obtained by averaging this per-index target over the index posterior just computed:

$$v_h^*(x, t, y) = \sum_{i=1}^N \Pr(I = i \mid X_t = x, t, \tilde{Y} = y) \cdot \frac{x^i - x}{1 - t} = \sum_{i=1}^N w_i^{(h)}(x, t, y) \frac{x^i - x}{1 - t},$$

which is (3). $\square$

Proof of Proposition 8. Throughout, ρ_Y denotes the marginal density of Y and $s(y) := \nabla \log \rho_Y(y) \in \mathbb{R}^k$ its score at y , both well defined since ρ_Y is positive and C^2 near y .

Step 0: an exact identity via a tilted population measure. Define the h -tilted density of Y ,

$$q_h(y') := \frac{K_h(y - y') \rho_Y(y')}{\int K_h(y - y'') \rho_Y(y'') dy''},$$

and, for a measurable f , $\mathbb{E}_h[f(X, Y)] := \mathbb{E}[K_h(y - Y)f(X, Y)]/\mathbb{E}[K_h(y - Y)]$, i.e. expectation under $Y \sim q_h$ with $X \mid Y$ unchanged. As $N \rightarrow \infty$ the empirical kernel-weighted moments $\bar{x}_h(y)$, $\text{Cov}_h(y)$ of Proposition 5 converge to their population counterparts $\mathbb{E}_h[X]$, $\text{Cov}_h^{\text{pop}}(y) := \text{Cov}_h[X]$ (a law of large numbers for the weighted sums $\sum_j K_h(y - y^j)x^j$ and $\sum_j K_h(y - y^j)$ separately, then Slutsky's theorem for the ratio); the discrepancy between the finite- N empirical moments and this population limit is the $O_P((Nh^k)^{-1/2})$ term addressed in Step 3. It therefore suffices to show

$$\text{Cov}_h^{\text{pop}}(y) = \Sigma(y) + h^2 J(y) J(y)^\top + O(h^4). \quad (12)$$

By the law of total covariance applied to X given Y under the tilted law $Y \sim q_h$,

$$\text{Cov}_h^{\text{pop}}(y) = \underbrace{\mathbb{E}_h[\Sigma(Y)]}_{\text{within-group}} + \underbrace{\text{Cov}_h[\mu(Y)]}_{\text{between-group}}, \quad (13)$$

which is an exact identity for every $h > 0$, since $\Sigma(Y) = \text{Cov}(X \mid Y)$ and $\mu(Y) = \mathbb{E}[X \mid Y]$ by definition and the law of total covariance requires no approximation.

Step 1: the tilted law of Y , rescaled. Substitute $y' = y + hu$ ($dy' = h^k du$) and write $K_h(y - y') = h^{-k} \varphi(u)$ with $\varphi(u) = (2\pi)^{-k/2} e^{-|u|^2/2}$ the standard Gaussian density (immediate from the definition of K_h). Then

$$\int K_h(y - y'') \rho_Y(y'') dy'' = \int \varphi(u) \rho_Y(y + hu) du,$$

and by the C^2 Taylor expansion $\rho_Y(y + hu) = \rho_Y(y)[1 + h s(y)^\top u + O(h^2)]$ (uniformly over the region where φ has non-negligible mass, using C^2 regularity of ρ_Y near y), together with $\int \varphi(u)u du = 0$,

$$\int \varphi(u)\rho_Y(y + hu) du = \rho_Y(y)[1 + O(h^2)].$$

Hence the tilted law of $u := (Y - y)/h$ under q_h has density

$$\tilde{q}_h(u) = \frac{\varphi(u)\rho_Y(y + hu)}{\int \varphi(u')\rho_Y(y + hu') du'} = \varphi(u)[1 + h s(y)^\top u + O(h^2)]. \quad (14)$$

Using $\mathbb{E}_\varphi[u] = 0$, $\mathbb{E}_\varphi[uu^\top] = I_k$, $\mathbb{E}_\varphi[u_a u_b u_c] = 0$ (odd moments of a centred Gaussian vanish) and $\mathbb{E}_\varphi[u_a u_b u_c u_d] = \delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}$, (14) gives the moments

$$\mathbb{E}_{\tilde{q}_h}[u] = \mathbb{E}_\varphi[u] + h \mathbb{E}_\varphi[u(s^\top u)] + O(h^2) = h \mathbb{E}_\varphi[uu^\top]s + O(h^2) = h s(y) + O(h^2), \quad (15)$$

$$\mathbb{E}_{\tilde{q}_h}[uu^\top] = \mathbb{E}_\varphi[uu^\top] + h \mathbb{E}_\varphi[uu^\top(s^\top u)] + O(h^2) = I_k + 0 + O(h^2), \quad (16)$$

the vanishing middle term in (16) because $\mathbb{E}_\varphi[u_a u_b u_c] = 0$ for every a, b, c . Consequently

$$\text{Cov}_{\tilde{q}_h}[u] = \mathbb{E}_{\tilde{q}_h}[uu^\top] - \mathbb{E}_{\tilde{q}_h}[u]\mathbb{E}_{\tilde{q}_h}[u]^\top = I_k + O(h^2) - h^2 s(y)s(y)^\top = I_k + O(h^2). \quad (17)$$

Step 2: within-group term. By C^1 smoothness of $\Sigma(\cdot)$, $\Sigma(y + hu) = \Sigma(y) + h[\nabla \Sigma(y) \cdot u] + O(h^2)$ (a matrix-valued first-order expansion, contracting the gradient 3-tensor with u), so by (14) and (15),

$$\mathbb{E}_h[\Sigma(Y)] = \mathbb{E}_{\tilde{q}_h}[\Sigma(y + hu)] = \Sigma(y) + h[\nabla \Sigma(y) \cdot \mathbb{E}_{\tilde{q}_h}[u]] + O(h^2) = \Sigma(y) + O(h^2),$$

since $\mathbb{E}_{\tilde{q}_h}[u] = O(h)$ by (15), making the second term $O(h^2)$ outright. This is the within-group term of (7).

Step 3: between-group term. By the C^2 Taylor expansion of μ ,

$$\mu(y + hu) = \mu(y) + hJ(y)u + \frac{h^2}{2}Q(u) + O(h^3), \quad Q(u)_a := \sum_{b,c} \partial_{y_b y_c}^2 \mu_a(y) u_b u_c,$$

and since covariance is invariant to the additive constant $\mu(y)$,

$$\begin{aligned} \text{Cov}_h[\mu(Y)] &= \text{Cov}_{\tilde{q}_h}\left[hJ(y)u + \frac{h^2}{2}Q(u) + O(h^3)\right] \\ &= h^2 J(y) \text{Cov}_{\tilde{q}_h}[u] J(y)^\top + h^3 \left[J(y) \text{Cov}_{\tilde{q}_h}(u, Q(u)) + \text{sym} \right] + h^4 \text{Cov}_{\tilde{q}_h}[Q(u)]. \end{aligned} \quad (18)$$

The leading term is $h^2 J(y)[I_k + O(h^2)]J(y)^\top = h^2 J(y)J(y)^\top + O(h^4)$ by (17). It remains to show the middle, apparently $O(h^3)$, cross term is in fact $O(h^4)$: this holds because $\text{Cov}_{\tilde{q}_h}(u, Q(u))$ itself vanishes at $h = 0$ and is only $O(h)$, not $O(1)$. Indeed, since $Q(u)$ is a quadratic form in u , $\mathbb{E}_{\tilde{q}_h}[u Q(u)^\top]$ and $\mathbb{E}_{\tilde{q}_h}[u] \mathbb{E}_{\tilde{q}_h}[Q(u)]^\top$ both involve only the third moments $\mathbb{E}_{\tilde{q}_h}[u_a u_b u_c]$ of $\tilde{q}_h$ (for the second factor, through $\mathbb{E}_{\tilde{q}_h}[u] = O(h)$ paired with $\mathbb{E}_{\tilde{q}_h}[Q(u)] = O(1)$, which is itself an $O(h)$ quantity overall); expanding (14) to the relevant order,

$$\begin{aligned} \mathbb{E}_{\tilde{q}_h}[u_a u_b u_c] &= \mathbb{E}_\varphi[u_a u_b u_c] + h \sum_e s_e(y) \mathbb{E}_\varphi[u_a u_b u_c u_e] + O(h^2) \\ &= 0 + h[s_c(y)\delta_{ab} + s_b(y)\delta_{ac} + s_a(y)\delta_{bc}] + O(h^2), \end{aligned}$$

using $\mathbb{E}_\varphi[u_a u_b u_c] = 0$ and the fourth-moment identity above. Thus every third moment of $\tilde{q}_h$ is itself $O(h)$ (it is exactly 0 under the untilted Gaussian φ), so $\text{Cov}_{\tilde{q}_h}(u, Q(u)) = O(h)$, and the cross term in (18) is $h^3 \cdot O(h) = O(h^4)$, absorbed into the stated $O(h^4)$. The final term $h^4 \text{Cov}_{\tilde{q}_h}[Q(u)] = O(h^4)$ trivially. Combining, $\text{Cov}_h[\mu(Y)] = h^2 J(y)J(y)^\top + O(h^4)$, the between-group term of (7).

Substituting Steps 2–3 into (13) proves (12), i.e. the deterministic part of (7).

Step 4: Monte-Carlo fluctuation. Write $S_N := \frac{1}{N} \sum_j K_h(y - y^j)$ and $T_N := \frac{1}{N} \sum_j K_h(y - y^j) x^j$, so $\bar{x}_h(y) = T_N/S_N$. Each is an average of N i.i.d. terms; a standard kernel-density computation gives $\text{Var}(K_h(y - Y)) = \Theta(h^{-k})$ (the kernel has height $\Theta(h^{-k})$ on an effective support of

Lebesgue measure $\Theta(h^k)$, so its second moment is $\Theta(h^{-k}) \cdot \Theta(h^k) \cdot \Theta(h^{-k}) = \Theta(h^{-k})$, dominating the squared mean $\Theta(1)$, and likewise for $K_h(y - Y)X$. Hence $S_N = \mathbb{E}[K_h(y - Y)] + O_P(\sqrt{h^{-k}/N})$ and $T_N = \mathbb{E}[K_h(y - Y)X] + O_P(\sqrt{h^{-k}/N})$, i.e. each has *relative* fluctuation $O_P((Nh^k)^{-1/2})$ around a mean of order $\Theta(1)$. A first-order Taylor (Slutsky) expansion of the ratio T_N/S_N around the population ratio $\mathbb{E}_h[X]$ then gives $\bar{x}_h(y) = \mathbb{E}_h[X] + O_P((Nh^k)^{-1/2})$, exactly as in standard Nadaraya–Watson consistency theory; the same argument applied to the (also ratio-of-averages) quadratic functional $\text{Cov}_h(y)$ gives $\text{Cov}_h(y) = \text{Cov}_h^{\text{pop}}(y) + O_P((Nh^k)^{-1/2})$, which is the remaining term of (7) and completes the proof. $\square$

Proposition 12 (Nadaraya–Watson rates for the conditional mean). *Under the assumptions of Proposition 8, $\bar{x}_h(y)$ is exactly the Nadaraya–Watson estimator of $\mu(y)$, with*

$$\text{Bias} = O(h^2), \quad \text{Var} = O((Nh^k)^{-1}), \quad \text{MSE} = O(h^4) + O((Nh^k)^{-1}),$$

minimised at $h^ \asymp N^{-1/(k+4)}$. Queries at $y = y^i$ are design points: x^i receives the maximal weight $K_h(0)$, and $Nh^k \rightarrow \infty$ is exactly what makes its relative influence vanish; without it the result does not apply there.*

Proof of Proposition 7. Fix (x, t, y) with $t < 1$ and condition on $I = i$. Then $X_t = (1 - t)X_0 + tx^i + t\rho\xi$, so with $r := x - tx^i$ we have $r = (1 - t)X_0 + t\rho\xi$, a linear combination of the independent standard Gaussians X_0, ξ ; hence $r \sim \mathcal{N}(0, s_t^2 I_d)$ with $s_t^2 = (1 - t)^2 + t^2\rho^2$, and $p(x \mid I = i, t) = s_t^{-d} \varphi_1(r/s_t) = \varphi_{s_t}(x - tx^i)$. As in Proposition 4, given $I = i$ the interpolant depends only on (X_0, ξ) and $\tilde{Y} = y^i + h\varepsilon$ only on ε , so $X_t \perp \tilde{Y} \mid I = i$ and Bayes’ rule gives $\Pr(I = i \mid X_t = x, t, \tilde{Y} = y) \propto \varphi_{s_t}(x - tx^i) K_h(y - y^i) = w_i$, the i -independent factor $1/N$ cancelling as before.

For the per-index target, (X_0, ξ, r) are jointly Gaussian with $\text{Cov}(X_0, r) = (1 - t)I_d$ and $\text{Cov}(\xi, r) = t\rho I_d$, so $\mathbb{E}[X_0 \mid r] = \frac{1-t}{s_t^2}r$ and $\mathbb{E}[\xi \mid r] = \frac{t\rho}{s_t^2}r$. With $U = X_1 - X_0 = x^i + \rho\xi - X_0$,

$$\mathbb{E}[U \mid X_t = x, t, I = i] = x^i + \rho \frac{t\rho}{s_t^2}r - \frac{1-t}{s_t^2}r = x^i + \frac{t\rho^2 - (1-t)}{s_t^2}(x - tx^i),$$

and averaging over the index posterior w_i gives, by Proposition 1,

$$v_h^*(x, t, y) = \sum_i w_i(x, t, y) \left[x^i + \frac{t\rho^2 - (1-t)}{s_t^2}(x - tx^i) \right], \quad w_i \propto \varphi_{s_t}(x - tx^i) K_h(y - y^i), \quad (19)$$

with φ_s the $\mathcal{N}(0, s^2 I_d)$ density.

For the endpoint law, argue exactly as in Lemma 3 and Theorem 1: consider the coupling $I \sim p^{(h)}(y)$, $X_0 \sim \pi_0$, $\xi \sim \mathcal{N}(0, I_d)$ mutually independent, $X_1 := x^I + \rho\xi$, with the same linear interpolant. Its index posterior at (x, t) is $\propto p_i^{(h)}(y) \varphi_{s_t}(x - tx^i)$, the same proportionality in i as w_i , so its conditional velocity is v_h^* ; the weights are smooth and positive and sum to one, so Lemma 2 applies and the flow transports π_0 to $\text{Law}(X_t)$ for every $t < 1$. Since $X_t = (1 - t)X_0 + tX_1 \rightarrow X_1$ almost surely as $t \uparrow 1$, $\text{Law}(X_t) \Rightarrow \text{Law}(x^I + \rho\xi) = \sum_i p_i^{(h)}(y) \mathcal{N}(x^i, \rho^2 I_d)$, which is (6). Its mean is $\sum_i p_i^{(h)} x^i = \bar{x}_h(y)$ and, by the law of total covariance applied to the mixture, its covariance is $\text{Cov}_h(y) + \rho^2 I_d$. Absolute continuity is immediate since each component is; (5) was derived from $\text{supp} \subseteq \{x^i\}$ and so does not apply. Finally $s_1^2 = \rho^2 > 0$, so $t \mapsto (t\rho^2 - (1 - t))/s_t^2$ is continuous at $t = 1$ (with value 1) and v_h^* is bounded on bounded sets up to and including $t = 1$. $\square$

Proof of Proposition 12. The bias is the $O(h^2)$ term identified in the proof of Proposition 8 (Taylor-expanding $\mu(y^j)$ about y against the symmetric kernel weights); squaring gives $\text{Bias}^2 = O(h^4)$. Averaging over an effective sample size $\asymp Nh^k$ gives the usual Nadaraya–Watson variance rate $O((Nh^k)^{-1})$. Summing, $\text{MSE}(h) = c_1 h^4 + c_2 (Nh^k)^{-1} + o(\cdot)$ for constants $c_1, c_2 > 0$; differentiating and setting to 0, $4c_1 h^3 = c_2 k N^{-1} h^{-k-1}$, i.e. $h^{k+4} \propto N^{-1}$, so $h^* \asymp N^{-1/(k+4)}$. $\square$

Proof of Proposition 9. Given $I = i$, write $r := x - tx^i$. Since $X_t = (1 - t)X_0 + tx^i + \gamma(t)Z$, $r = (1 - t)X_0 + \gamma(t)Z$, a linear combination of the independent standard Gaussians $X_0, Z \sim \mathcal{N}(0, I_d)$, so $r \sim \mathcal{N}(0, s_t^2 I_d)$ with $s_t^2 = (1 - t)^2 + \gamma(t)^2$. For jointly Gaussian (X_0, r) , $\text{Cov}(X_0, r) = (1 - t)I_d$

(the $\gamma(t)Z$ term is independent of X_0), so the standard Gaussian conditional-mean formula gives $\mathbb{E}[X_0 | r] = \text{Cov}(X_0, r) \text{Var}(r)^{-1} r = \frac{1-t}{s_t^2} r$; identically, $\text{Cov}(Z, r) = \gamma(t)I_d$ gives $\mathbb{E}[Z | r] = \frac{\gamma(t)}{s_t^2} r$. Hence

$$v_\sigma^*(x, t, y^i) = \mathbb{E}[\dot{X}_t | X_t = x, t, I = i] = x^i - \mathbb{E}[X_0 | r] + \dot{\gamma}(t) \mathbb{E}[Z | r] = x^i + \frac{-(1-t) + \dot{\gamma}(t)\gamma(t)}{s_t^2} r.$$

Since $\gamma(t)^2 = \sigma^2 t(1-t)$, differentiating gives $2\dot{\gamma}\gamma = \sigma^2(1-2t)$, i.e. $\dot{\gamma}\gamma = \frac{\sigma^2}{2}(1-2t)$, which yields (a)'s expression for $c(t)$. For the logarithmic-derivative identity, $\frac{d}{dt}s_t^2 = -2(1-t) + 2\dot{\gamma}\gamma = 2s_t^2 c(t)$ by the same computation, so $c(t) = \frac{1}{2} \frac{d}{dt} \log s_t^2$.

(b) Along the flow, $u_t := x_t - tx^i$ satisfies $\dot{u}_t = \dot{x}_t - x^i = v_\sigma^*(x_t, t, y^i) - x^i = c(t)u_t$ by (a) (with $r = u_t$ evaluated on the trajectory), a linear ODE with solution $u_t = u_0 \exp(\int_0^t c(\tau) d\tau) = x_0 \exp(\frac{1}{2}[\log s_t^2 - \log s_0^2]) = x_0 s_t$, using $s_0 = 1$ and $u_0 = x_0 - 0 \cdot x^i = x_0$. Hence $x_t = tx^i + s_t x_0$.

(c) Since $s_1 = (1-1)[(1-1) + \sigma^2 \cdot 1] = 0$, (b) gives $x_1 = x^i$ for every $x_0 \sim \pi_0$ and every $\sigma \geq 0$, so $p_1^{\text{cond}}(\cdot | y^i) = \delta_{x^i}$. $\square$

Corollary 5 (What interpolant noise does and does not change). *Interpolant noise does not change the population endpoint law (9) for any σ ; it does raise the irreducible error ($\inf_v L_\sigma > 0$ for $\sigma > 0$, whereas $\inf_v L_0 = 0$) and halves the Lipschitz blow-up rate: $\lim_{t \uparrow 1} (1-t)|c(t)| = 1$ for $\sigma = 0$ and $\frac{1}{2}$ for $\sigma > 0$.*

Proof of Corollary 5. Endpoint invariance is Proposition 9(c).

Irreducible error. For $\sigma = 0$, $\gamma \equiv 0$ so $\dot{X}_t = X_1 - X_0$ is exactly the target of Proposition 2, giving $\inf_v L_0 = 0$. For $\sigma > 0$: by the same projection argument as Proposition 1, applied to $\dot{X}_t$ and (X_t, t, Y) , it suffices to show $\text{Var}(\dot{X}_t | X_t = x, t, I = i) > 0$ for some positive-measure set of t . Writing $\dot{X}_t - x^i = -X_0 + \dot{\gamma}(t)Z$ and $r = (1-t)X_0 + \gamma(t)Z$ as in the proof of Proposition 9, the standard Gaussian conditioning formula gives, for $u = (-1, \dot{\gamma}(t))$ and $w = (1-t, \gamma(t))$,

$$\text{Var}(\dot{X}_t | r, I = i) = |u|^2 - \frac{(u \cdot w)^2}{|w|^2} = (1 + \dot{\gamma}(t)^2) - c(t)^2 s_t^2,$$

using $u \cdot w = -(1-t) + \dot{\gamma}(t)\gamma(t) = c(t)s_t^2$ from the proof of Proposition 9. This vanishes only if $u \parallel w$, i.e. $\dot{\gamma}(t)(1-t) = -\gamma(t)$; substituting $\gamma(t) = \sigma\sqrt{t(1-t)}$ and $\dot{\gamma}(t) = \sigma(1-2t)/(2\sqrt{t(1-t)})$ reduces this to $1-2t = -2t$, i.e. $1 = 0$ —impossible. So for $\sigma > 0$ the residual variance is strictly positive for every $t \in (0, 1)$, and $\inf_v L_\sigma(v) = \mathbb{E}[\text{Var}(\dot{X}_t | X_t, t, Y)] > 0$.

Lipschitz rate. From (a), $(1-t)c(t) = \frac{-(1-t) + \frac{\sigma^2}{2}(1-2t)}{(1-t) + \sigma^2 t}$, which as $t \uparrow 1$ tends to -1 if $\sigma = 0$ and to $-\sigma^2/2 / \sigma^2 = -\frac{1}{2}$ if $\sigma > 0$, giving the stated limits. $\square$

Proof of Proposition 10. By Theorem 1, for $h > 0$ (and $\sigma = 0$) the endpoint law $p_h^{\text{gen}}(\cdot | y) = \sum_i p_i^{(h)}(y) \delta_{x^i}$ depends on h only through the label factor $K_h(y - y^i)$ in (3), and Corollaries 2–3 show it genuinely interpolates between two distinct endpoint laws as h ranges over $[0, \infty)$. By contrast, for $h = 0$ with distinct labels and any $\sigma \geq 0$, $Y = y^i$ already pins $I = i$ deterministically, so no index-posterior argument is needed: Proposition 9 shows directly that although the spatial factor of Lemma 1, $\mathcal{N}(tx^i, (1-t)^2 I_d)$, is replaced by $\mathcal{N}(tx^i, s_t^2 I_d)$ for every i identically, the endpoint is exactly invariant, $p_1^{\text{cond}}(\cdot | y^i) = \delta_{x^i}$ for every σ . Since perturbing the label factor changes p_1 while perturbing the spatial factor identically across i does not, the two mechanisms act through different, non-interchangeable channels. $\square$

Proof of Proposition 11. Let X, X' be i.i.d. copies of $X_t | I = i \sim \mathcal{N}(tx^i, (1-t)^2 I_d)$ and $e(\cdot) = f_i(\cdot, t) - v(\cdot, t, y^i)$. Since $f_i(X, t) - f_i(X', t) = -(X - X')/(1-t)$ and $X - X' \sim \mathcal{N}(0, 2(1-t)^2 I_d)$,

$$\mathbb{E}|f_i(X, t) - f_i(X', t)|^2 = \frac{2(1-t)^2 d}{(1-t)^2} = 2d.$$

By L -Lipschitzness, $\mathbb{E}|v(X, t, y^i) - v(X', t, y^i)|^2 \leq L^2 \mathbb{E}|X - X'|^2 = 2dL^2(1-t)^2$. Also $\mathbb{E}|e(X) - e(X')|^2 = 2\mathbb{E}|e(X)|^2 - 2|\mathbb{E}e(X)|^2 \leq 2\mathbb{E}|e(X)|^2$ (since X, X' are i.i.d.). Applying the $L^2(\Omega)$ triangle inequality to $f_i(X, t) - f_i(X', t) = [e(X) - e(X')] + [v(X, t, y^i) - v(X', t, y^i)]$,

$$\sqrt{2d} \leq \sqrt{2\mathbb{E}|e(X)|^2} + \sqrt{2d}L(1-t),$$

hence $\sqrt{\mathbb{E}|e(X)|^2} \geq \sqrt{d}[1-L(1-t)]_+$, i.e. $\mathbb{E}|e(X)|^2 \geq d[1-L(1-t)]_+^2$, which is (10)'s pointwise bound with $e(X) = v(X_t, t, y^i) - f_i(X_t, t)$ up to sign. (Using the sharp identity $\mathbb{E}|e(X) - e(X')|^2 = 2\mathbb{E}|e(X)|^2 - 2|\mathbb{E}e(X)|^2 \leq 2\mathbb{E}|e(X)|^2$, rather than the cruder bound $4\mathbb{E}|e(X)|^2$, is what preserves the constant d in (10); a weaker triangle-inequality step would give only a smaller floor.) $\square$

Proof of Corollary 4. Substituting $s = 1 - t$, $\int_0^{1/L} (1 - Ls)^2 ds = [s - Ls^2 + \frac{L^2 s^3}{3}]_0^{1/L} = \frac{1}{L} - \frac{1}{L} + \frac{1}{3L} = \frac{1}{3L}$. Proposition 11's bound (10)'s integrand holds for every training index i , hence for every $v \in \mathcal{F}_L$ and every $t \in [1 - 1/L, 1)$, so it survives averaging over $I \sim \text{Unif}\{1, \dots, N\}$ and $t \sim \text{Unif}(0, 1)$, giving $\inf_{v \in \mathcal{F}_L} L_0(v) \geq d \int_{1-1/L}^1 [1 - L(1-t)]^2 dt = d/(3L)$. $\square$

B FINITE-SAMPLE MEMORISATION AND EXACT NON-REPRESENTABILITY

This subsection translates two representability results from the unconditional gradient-variance/stochastic-interpolant literature (Reu et al., 2025) into the present notation: (i) why a fixed finite *unconditional* batch already admits a zero-loss memorising field despite Proposition 3's positive population loss, and (ii) why no finite network can exactly represent the collapsed field $(\star)$. Unlike Sections 3.1–3.5, the finite sample itself is treated as random here.

Lemma 4 (Generic non-intersection of finite interpolant segments). *Let $x_0^1, \dots, x_0^N \stackrel{\text{iid}}{\sim} \pi_0$ and, independently, $z^1, \dots, z^N \stackrel{\text{iid}}{\sim} \rho_1$ be N unconditional source–target pairs (no labels), both laws absolutely continuous on $\mathbb{R}^d$, and $\ell_i(t) = (1-t)x_0^i + tz^i$, $t \in [0, 1]$. (a) If $d \geq 2$: for any fixed $i \neq j$, $\Pr[\exists t \in (0, 1) : \ell_i(t) = \ell_j(t)] = 0$. (b) If $d > 2$: for any fixed $i \neq j$, $\Pr[\exists t_i, t_j \in (0, 1), t_i \neq t_j : \ell_i(t_i) = \ell_j(t_j)] = 0$.*

Proof. (a) Fix $\hat{t} \in (0, 1)$; $\ell_i(\hat{t}) = \ell_j(\hat{t})$ solves to $z^j = \frac{1-\hat{t}}{\hat{t}}(x_0^i - x_0^j) + z^i$, which pins z^j to a one-dimensional affine subspace of $\mathbb{R}^d$ determined by $(x_0^i, x_0^j, z^i, \hat{t})$. This subspace is Lebesgue-null whenever $d \geq 2$, and ρ_1 is absolutely continuous, so conditionally on (x_0^i, x_0^j, z^i) the event has probability 0; integrating over (x_0^i, x_0^j, z^i) preserves probability 0, and the bound is uniform in $\hat{t}$, so the same holds for "some $t = \hat{t}$ " (the constraint is on z^j alone and does not depend on which $\hat{t}$ occurs). A union bound over the finitely many pairs (i, j) finishes (a). (b) With $t_i \neq t_j$ both free, solving for z^j pins it to a two-dimensional affine subspace, Lebesgue-null when $d > 2$; the rest of the argument is identical. $\square$

Proposition 13 (A memorising field always exists on a fixed finite unconditional batch). *Fix N unconditional pairs $\{(x_0^i, z^i)\}_{i=1}^N$ as in Lemma 4 with $d > 2$ and the z^i pairwise distinct a.s., and sample m time points $t^{(i,j)} \in (0, 1)$, $j = 1, \dots, m$, independently across (i, j) . Let $X_t^{(i,j)} = (1 - t^{(i,j)})x_0^i + t^{(i,j)}z^i$ and*

$$L_{\text{MC}}^{\text{unc}}(v) = \frac{1}{Nm} \sum_{i=1}^N \sum_{j=1}^m |(z^i - x_0^i) - v(X_t^{(i,j)}, t^{(i,j)})|^2.$$

Then, almost surely over the draw of $\{(x_0^i, z^i)\}$ and $\{t^{(i,j)}\}$, some v attains $L_{\text{MC}}^{\text{unc}}(v) = 0$ —even though $\inf_v L_{\text{unc}}(v) > 0$ at the population level whenever the z^i are not all equal (Proposition 3).

Proof. By Lemma 4(a)–(b), union-bounded over the $\binom{N}{2}$ pairs, almost surely no two of the N full segments $\{\ell_i(t) : t \in (0, 1)\}$ intersect at all, so in particular the Nm sampled points $(X_t^{(i,j)}, t^{(i,j)})$ never collide across different i . On this full-measure event $(z, t) \mapsto i$ is well defined on the sample, so setting $v(z, t) := z^{i(z,t)} - x_0^{i(z,t)}$ on the sample and $v := 0$ elsewhere gives $v(X_t^{(i,j)}, t^{(i,j)}) = z^i - x_0^i$ exactly for every sampled (i, j) , so every summand of $L_{\text{MC}}^{\text{unc}}$ vanishes. $\square$

Remark 1 (Why SGD does not land here). Proposition 13 is valid only for one *fixed* finite sample. Standard CFM training resamples x_0^i (and the time samples) independently every epoch, so the lookup table fit at epoch k is checked, at epoch $k + 1$, against points it never saw; no single lookup table stays consistent with a growing stream of (x_0, t) draws. This is the precise mechanism behind the informal claim that resampling x_0 —not mere continuity of π_0 —is what pushes unconditional training toward the population optimum of Proposition 3 rather than a per-example memorising solution, and it gives a concrete reading of P6 (Table 6): at fixed capacity, larger N makes the epoch-consistent lookup table harder to realise with a smooth finite-parameter function, pushing the attained solution back toward the population optimum. Note the asymmetry with Proposition 2: conditional collapse needs no genericity argument and no fixed-batch caveat, because y^i pins down i deterministically for every x_0 , not merely with probability 1 on one finite draw—resampling x_0 every epoch does nothing to prevent it, which is exactly why conditional collapse is universal while unconditional memorisation is not.

Lemma 5 (The collapse field cannot be exactly represented by a finite network). *Let $g(t) = 1/(1-t)$ on $[0, 1)$ and let $f_\theta : \mathbb{R} \rightarrow \mathbb{R}$ be any finite-depth, finite-width feedforward network with finite real weights, built from affine layers and globally Lipschitz activations (this covers ReLU, leaky-ReLU, tanh, sigmoid, GELU/SiLU, softplus, and every activation used in our experiments). Then $\sup_{t \in [0, 1)} |f_\theta(t) - g(t)| = \infty$.*

Proof. A finite composition of affine maps (finite weight matrices, hence finite operator norm) and globally Lipschitz activations is globally Lipschitz on $\mathbb{R}$, with constant equal to the product of the finitely many layer constants. A Lipschitz function is bounded on any bounded set: fixing $t_0 \in [0, 1)$, $|f_\theta(t)| \leq |f_\theta(t_0)| + K|t - t_0| \leq |f_\theta(t_0)| + K$ for all $t \in [0, 1)$, so $\sup_{t \in [0, 1)} |f_\theta(t)| < \infty$. But $g(t) \rightarrow \infty$ as $t \uparrow 1$, so g is unbounded on $[0, 1)$, hence so is $f_\theta - g$. $\square$

For $x \neq x^i$ fixed, $t \mapsto v_0^*(x, t, y^i) = (x^i - x)/(1 - t)$ has exactly this $1/(1 - t)$ blow-up along the direction $x^i - x$, so Lemma 5 (applied coordinatewise to that direction) shows that *no* finite network of the kind used in Section 4 can represent v_0^* *exactly* on all of $[0, 1)$: for every fixed parameter vector θ there is some (x, t) at which $v_\theta \neq v_0^*$. This does not contradict universal approximation, which guarantees only *uniform* approximation on *compact* sets: every $[0, 1 - \delta]$, $\delta > 0$, is disjoint from the singularity at $t = 1$ —exactly the regime the sampler must traverse to reach the collapse point x^i , and exactly where the ODE solver’s $t = 1$ handling matters in practice. We stop short, however, of claiming this pointwise non-representability forces $L_{\text{model}}^* - L_{\text{all}}^* > 0$ for an architecture whose weights (hence Lipschitz constant) are unconstrained: pointwise blow-up on a shrinking neighbourhood of $t = 1$ is compatible, in principle, with a sequence of increasingly ill-conditioned finite networks driving the *expected* loss $L_0(v_\theta)$ arbitrarily close to 0 (consistent with the growing empirical Lipschitz constant we measure in Figure 5). The genuine, quantitative loss floor we prove, $d/(3L)$, therefore applies only to the Lipschitz-capped class $\mathcal{F}_L$ of Corollary 4, and should not be read as a statement about arbitrary finite architectures.

C ADDITIONAL EXPERIMENTAL RESULTS

C.1 ARCHITECTURE AND OPTIMISER SENSITIVITY

Every headline result in Section 4 uses one fixed architecture (4-layer, width-128 MLP, Adam). Since Proposition 2 constrains only the *population* minimiser, not any parameterisation, the theory predicts collapse for *any* sufficiently expressive architecture; what it does not predict is how fast a given architecture and optimiser reach it within a fixed iteration budget. We re-ran EXP-1 (conditional, 2 seeds each) varying one factor at a time—width $\in \{32, 64, 256\}$ at depth 4, depth $\in \{2, 6\}$ at width 128, and the optimiser (SGD with momentum 0.9, $\text{lr} = 10^{-2}$, in place of Adam)—holding everything else, including the 2×10^5 -iteration budget, fixed.

Table 2 reports the result. None of these variants contradicts Proposition 2: all seven configurations collapse monotonically from the same $\approx \text{tr } \Sigma_{\text{post}}$ starting point (not shown), and none plateaus above the baseline’s level by more than a factor of ~ 2 —exactly the qualitative picture the theory predicts (every sufficiently expressive architecture converges toward the same population optimum) together with the optimisation-gap caveat of Section 5 (how close a given budget gets you depends

on the architecture and optimiser). We did not extend this sweep to the Lipschitz diagnostic of Appendix C.9 or to EXP-2/EXP-3; doing so is left to future work.

configuration	width	depth	optimiser	tr Cov at 2×10^5
baseline	128	4	Adam	0.396 ± 0.127
narrower	32	4	Adam	0.657 ± 0.058
	64	4	Adam	0.526 ± 0.058
wider	256	4	Adam	0.403 ± 0.079
shallower	128	2	Adam	0.860 ± 0.156
deeper	128	6	Adam	0.407 ± 0.127
SGD	128	4	SGD(m=0.9)	0.711 ± 0.193

Table 2: Architecture/optimiser ablation on EXP-1 (2 seeds each), same 2×10^5 -iteration budget as Table 1. Collapse speed increases with capacity but saturates beyond roughly the baseline’s size (width 256 and depth 6 both land within seed noise of the width-128/depth-4 baseline, despite reaching a strictly lower training loss—0.46 vs. 0.59 for width 256—so more capacity buys a better loss without a proportionally deeper collapse at this budget); under-capacity architectures (width 32, depth 2) collapse markedly less within the same budget; and the optimiser matters substantially, with SGD reaching only about half the collapse Adam does here.

C.2 ADVERSARIAL PAIRING, IN FULL

C.3 ADVERSARIAL PAIRING: COLLAPSE DOES NOT REQUIRE A REAL POSTERIOR

Proposition 2 never uses A or σ_{obs} : it needs only that $y^1, \dots, y^N$ be pairwise distinct. It therefore predicts that collapse is unchanged if Y is *randomly permuted relative to X* before training, so that (x^i, y^i) is no longer the true generative pair—identification, not discovered posterior structure. This mirrors the “Adversarial Pairings” experiment of Reu et al. (2025) (shuffled OT-coupling targets on CelebA, unconditional), adapted to the conditional case where the prediction is exact. Re-running EXP-1 unchanged (5 seeds), permuting Y independently per seed, tr Cov collapses under shuffled labels just as under the real pairing, from $\gtrsim 1$ to 0.625 ± 0.108 at 2×10^5 (vs. 0.396 ± 0.127 real); the gap is optimisation difficulty at a fixed budget—a random x - y pairing is a harder target to fit—not a difference in the population optimum, which Proposition 2 fixes at 0 either way. The sharper diagnostic is $\|\text{mean} - \mu_{\text{post}}(y^i)\|$, the distance to y^i ’s *true* analytic posterior mean (still computable from A , but now unrelated to the x^i the label was shuffled onto): it does not shrink but *grows* throughout training, $0.84 \pm 0.13 \rightarrow 1.437 \pm 0.176$, crossing above $\|\text{mean} - x^i\|$ ($1.35 \pm 0.10 \rightarrow 0.559 \pm 0.104$) between 10^4 and 3×10^4 iterations and staying above it thereafter (Figures 13–14, appendix). The generated law tracks the arbitrary label-to-example assignment, not any statistical relationship between x and y —Corollary 1’s conclusion, confirmed even when the labels carry no real information about x at all.

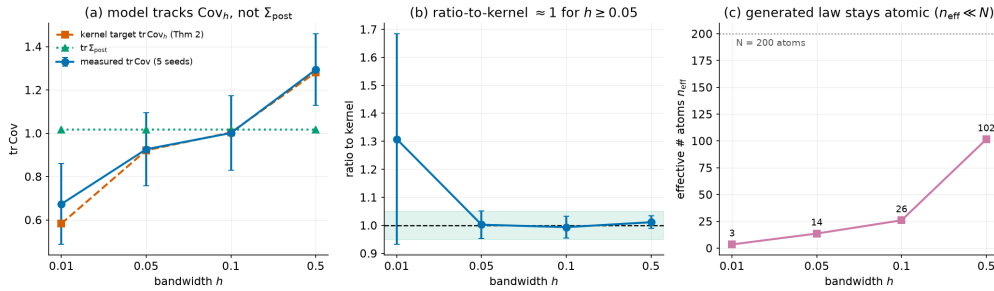


Figure 1: P7 headline (5 seeds, error bars over seeds). (a) Generated tr Cov follows the kernel target tr Cov_h (dashed) and crosses tr Σ_{post} (dotted) only incidentally near $h=0.1$. (b) Ratio-to-kernel stays in the shaded $[0.95, 1.05]$ band for $h \geq 0.05$. (c) $n_{\text{eff}} \ll N$: a sparse mixture of atoms (Proposition 6), not a continuous posterior.

C.4 THE ATOMICITY FLOOR, MEASURED

Table 3 evaluates the *population* endpoint laws of Theorem 1 and Proposition 7 directly, so it bounds what any trained model could achieve and is independent of optimisation. For each (d, N) we draw the analytic posterior at six training labels, compute the floor $F = \mathbb{E}_{x \sim p(\cdot|y)} \text{dist}(x, \{x^i\})^2$ of (5) exactly, and report the best MMD to that posterior over h (atomic, $\rho = 0$) and over (h, ρ) (endpoint-smoothed). Script: `scripts/atomicity_scaling.py`; the single-configuration sweep behind the numbers quoted in Section 4.2 is `scripts/verify_target_noise.py`.

d	N	$\text{tr } \Sigma_{\text{post}}$	floor F	$F / \text{tr } \Sigma_{\text{post}}$	MMD		entropic OT	
					atomic	gain from ρ	atomic	gain from ρ
2	1000	1.004	0.0073	0.007	0.0048	1.01×	3.33	1.17×
2	200	1.004	0.0353	0.035	0.0115	1.02×	16.75	1.31×
2	50	1.004	0.1932	0.192	0.0742	1.24×	60.41	1.57×
5	200	4.001	0.9242	0.231	0.0059	1.05×	277.96	1.31×
10	200	9.001	4.8303	0.537	0.0105	1.07×	1638.21	1.13×
10	50	9.001	7.0125	0.779	0.0434	1.18×	2406.63	1.16×

Table 3: The atomicity obstruction, quantified at the population optimum. F is the exact bound of (5); its relative size follows the $N^{-2/d}$ nearest-atom scaling. “Gain from ρ ” is the best score over h alone divided by the best over (h, ρ) . Two readings. (i) The OT gain exceeds the MMD gain at every row: an entropic OT divergence sees the atomic support, an RBF MMD at the median-heuristic bandwidth largely does not—which is why Figure 6’s MMD curves must not be read as measuring atomicity. (ii) At fixed $d=2$ the OT gain grows with the floor ($1.17 \rightarrow 1.31 \rightarrow 1.57\times$ as N falls), but *across* d it does not: $d=10$ has by far the largest floor and among the smallest gains, because an isotropic ρ cannot fill a high-dimensional gap without paying $d\rho^2$ in variance. Endpoint smoothing is defeated by dimension, exactly as kernel density estimation is. The OT column uses a coarser (h, ρ) grid and $M=400$, so its absolute values are not comparable to the $M=2000$ single-configuration sweep quoted in Section 4.2.

C.5 ENDPOINT SMOOTHING ON A TRAINED MODEL, PER SEED

Table 4 gives the per-seed numbers behind the trained-model claim of Section 4.2. Both arms use the EXP-1 architecture and schedule unchanged at $N=50$ and $h=0.1$, differing only in ρ (`configs/exp1_target_noise.yaml`, `train.target_noise_rho`); analysis by `scripts/analyze_target_noise.py`. The measured tr Cov tracks $\text{tr Cov}_h + d\rho^2$ per seed as well as on average, and the OT improvement is $2.5\times$ in each seed separately—the aggregate is not carried by one run. Seed variation is large in absolute terms (seed 0’s problem instance is markedly harder than seed 1’s), which is why the ratio, not the level, is the quantity to read.

arm	seed	tr Cov	predicted $\text{tr Cov}_h + d\rho^2$	MMD	entropic OT	$\ \text{mean} - \mu_{\text{post}}\ $
$\rho = 0$	0	0.2176	0.2108	0.2330	118.90	0.5176
	1	0.4353	0.4436	0.1168	39.45	0.2790
	mean	0.3265	0.3272	0.1749	79.18	0.3983
$\rho = 0.3$	0	0.3959	0.3908	0.1549	47.38	0.5383
	1	0.5870	0.6236	0.0798	15.70	0.2948
	mean	0.4914	0.5072	0.1174	31.54	0.4166

Table 4: Endpoint smoothing in a trained model ($N=50$, $h=0.1$, 2×10^5 iterations, 8 conditions, $M=800$). The generated covariance matches (6)’s $\text{tr Cov}_h + d\rho^2$ to within 3% in both arms, and the mean is unchanged by ρ —both as the moment formula requires. Distance to the analytic posterior falls $2.5\times$ in entropic OT but only $1.5\times$ in MMD, the same metric-sensitivity gap seen at the population level in Table 3.

C.6 HELD-OUT CONDITIONS, PER SEED

Table 5 gives the per-seed numbers behind the held-out paragraph of Section 4.1. Four of the five seeds behave alike—held-out tr Cov ends within a factor of ~ 2 of the training-condition value, i.e. the same collapse, onto the nearest memorised atom rather than onto the posterior of that y (Corollary 2). Seed 0 is the exception: its held-out ODE integration diverges between 10^5 and 2×10^5 iterations, which is why we report medians in the main text. We did not investigate whether this divergence is a property of that problem instance (the seed also draws A) or of the integrator’s fixed step count near $t=1$; distinguishing the two would need a step-size sweep we did not run.

group	per seed, tr Cov at 2×10^5					5-seed	
	0	1	2	3	4	mean	median
training conditions	0.407	0.547	0.450	0.417	0.162	0.396	0.417
held-out conditions	3.0×10^5	0.212	0.429	0.364	0.564	6.0×10^4	0.429

Table 5: Held-out vs. training conditions at 2×10^5 iterations (8 held-out y per run). Four seeds collapse on held-out labels much as on training labels; seed 0’s held-out integration diverges, making the 5-seed mean uninformative. Up to 10^5 iterations no seed diverges and the two groups agree closely (held-out 0.51 vs. training 0.58).

C.7 SCOPE-MARKED SWEEPS (P5, P6, d/k ; EMPIRICAL)

The following depend on N and σ_{obs} at fixed architecture and are *not* determined by the population theory (Section 3.7); we report them as empirical findings using the collapse ratio $\text{tr Cov} / \text{tr } \Sigma_{\text{post}}$ at 2×10^5 iterations (Tables 6–7 and Figure 2).

N	50	200	1000	5000	(d, k)	(2, 1)	(10, 1)	(10, 10)
ratio	0.049	0.389	0.912	0.990	ratio	0.390	0.317	0.0065
$\pm$	0.038	0.123	0.103	0.031	$\pm$	0.127	0.033	0.0024

(a) P6: collapse vs. dataset size.

(b) Collapse vs. dimension/observation.

Table 6: 5-seed sweeps (empirical). P6 is monotone and stable across seeds: small N collapses almost completely, $N=5000$ barely collapses ($\approx$ true posterior)—a representation/optimisation statement, not a contradiction of Proposition 2. Larger k (a more informative label) sharpens the identifier and deepens collapse, consistent with the identification mechanism.

σ_{obs}	0.01	0.1	0.5	1.0
$\text{tr } \Sigma_{\text{post}}$	1.000	1.018	1.266	1.531
generated tr Cov	0.472 ± 0.108	0.396 ± 0.127	0.416 ± 0.155	0.507 ± 0.236
collapse ratio	0.472	0.390	0.328	0.331

Table 7: P5 (empirical), 5 seeds. The between- σ_{obs} spread in the ratio (0.33–0.47, range ~ 0.14) is *smaller* than the between-seed std at each point (0.11–0.24): the sweep is flat within seed noise. This is consistent with an identification-through- y mechanism that operates for every $\sigma_{\text{obs}} > 0$; the theory implies no monotonicity, so flatness is not a falsification.

C.8 AN N -SWEEP ON EXP-3, AND WHAT IT ACTUALLY MEASURES (EMPIRICAL)

Table 6 sweeps N on EXP-1 at a fixed iteration budget. We repeat that sweep on the image experiment, and add the control the EXP-1 version lacks.

At $N \in \{100, 500, 2000\}$ and a fixed 30000 iterations with batch 64 (3 seeds each), the inpaint-region pixel variance rises monotonically with N , by $14.9\times$, with a log-log slope of $+0.90$ (Table 8A). Taken alone that looks like a clean scaling law in N . It is not safe to read it that way. At a

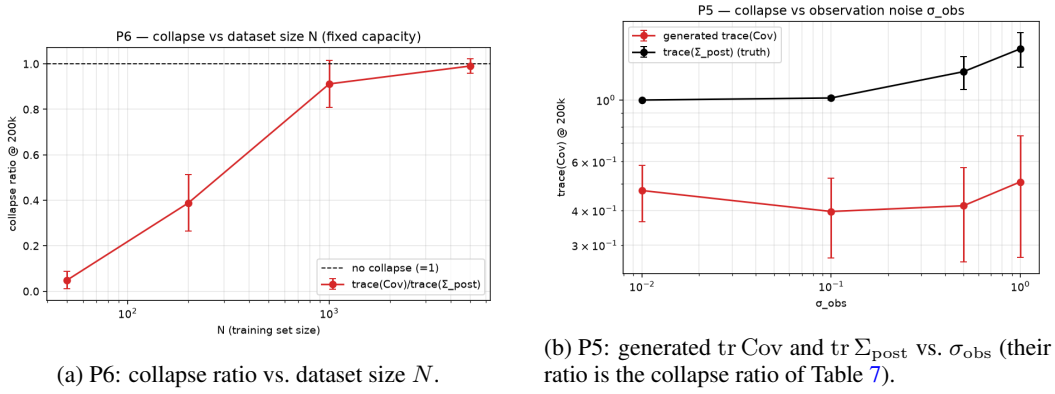


Figure 2: Scope-marked sweeps (5 seeds, error bars over seeds). Collapse deepens monotonically as N shrinks (a); it is flat within seed noise across σ_{obs} (b). Neither is determined by the population theory; both are reported as empirical findings (Section 3.7).

fixed budget the number of gradient samples each training image receives is $30000 \times 64/N$, i.e. exactly proportional to $1/N$, so “variance grows like N ” and “variance grows as optimisation progress per image shrinks” are the same curve. A slope of $\approx +1$ is what *either* explanation predicts.

Two controls separate them. Holding the per-image exposure fixed at 3840 gradient samples instead ($N=100$ at 6k iterations, $N=500$ at 30k, $N=2000$ at 120k), the slope drops to -0.29 and the spread from $14.9 \times$ to $2.3 \times$; $N=500$ and $N=2000$ then differ by 3%, against a 17% seed-to-seed standard deviation (Table 8B, Figure 3). Lining the runs up by training loss instead of by iteration—a control that does not assume exposure is the right currency—gives the same answer: $N=500$ and $N=2000$ agree to 6–13% with inconsistent sign at every loss level, while $N=100$ sits about $2 \times$ lower (Table 8C).

So most of the apparent N -dependence at fixed budget is an optimisation-budget effect, and the residual genuine effect is a factor ≈ 2 rather than ≈ 15 . This is what Proposition 2 would lead one to expect: the population minimiser is collapsed for *every* finite N , so a large intrinsic N -dependence would be the surprising outcome. It also sharpens the caveat already attached to the EXP-1 sweep in Table 6, which shares the same fixed-budget design and therefore the same confound: that sweep is best read as a statement about how far a fixed budget gets you at each N , not about how the population optimum depends on N . We have not rerun the EXP-1 sweep under an exposure control; the EXP-3 evidence here is what supports the reading.

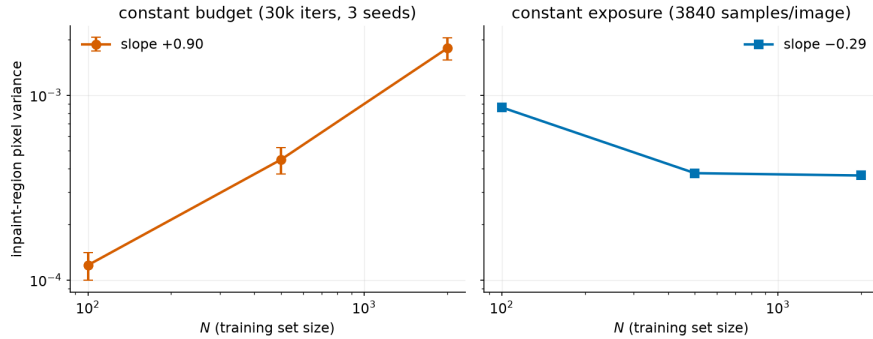


Figure 3: EXP-3 N -sweep on a shared y -axis. Left: at a fixed 30000-iteration budget the inpaint-region variance rises like $N^{+0.90}$. Right: at a fixed per-image gradient exposure the same three N values are nearly flat. The difference between the panels is the confound, not the data.

A. constant budget (30000 iters, 3 seeds)	$N=100$	$N=500$	$N=2000$
grad. samples per image	19200	3840	960
inpaint pixel variance	$1.20(21) \times 10^{-4}$	$4.48(74) \times 10^{-4}$	$1.79(24) \times 10^{-3}$
NN distance to train image	$7.4(2.9) \times 10^{-5}$	$3.71(5) \times 10^{-4}$	$1.30(31) \times 10^{-3}$
B. constant exposure (3840/image, seed 0)	$N=100$	$N=500$	$N=2000$
iterations	6000	30000	120000
inpaint pixel variance	8.59×10^{-4}	3.79×10^{-4}	3.68×10^{-4}
final training loss	0.0159	0.0089	0.0080
C. matched training loss	$N=100$	$N=500$	$N=2000$
at loss 0.020	1.09×10^{-3}	2.77×10^{-3}	3.13×10^{-3}
at loss 0.015	7.76×10^{-4}	1.23×10^{-3}	1.37×10^{-3}
at loss 0.010	3.15×10^{-4}	6.16×10^{-4}	5.80×10^{-4}

Table 8: EXP-3 N -sweep (empirical, not determined by the population theory). Panel A is the fixed-budget sweep: slope $+0.90$ in $\log N$, spread $14.9\times$. Panel B holds the per-image gradient exposure fixed instead: slope -0.29 , spread $2.3\times$. Panel C lines the runs up by training loss, interpolating over every checkpoint of every run at that N . Parenthesised digits are the standard deviation over 3 seeds in the last digits shown; panels B and C use a single seed per point, so the $N=100$ gap in B (whose run is also the least converged, loss 0.0159) is best read together with C rather than alone.

C.9 ADDITIONAL VERIFICATION (GAP TO OPTIMUM, LIPSCHITZ, POSTERIOR DISTANCE)

Three checkpoint-only measurements corroborate the theory. (i) *Optimality gap*. The Monte-Carlo gap $L(v_\theta) - L(v_h^*)$, with $L(v_h^*) = \mathbb{E}[\text{Var}(U \mid X_t, t, \tilde{Y})]$ computed from the exact minimiser, is non-negative at every checkpoint and decreases monotonically toward 0 at every h (e.g. $h=0.1$: $1.14 \rightarrow 0.22 \rightarrow 0.102$ at $10^2/3 \times 10^4/2 \times 10^5$ steps), consistent with ratio-to-kernel $\rightarrow 1$. The irreducible error $L(v_h^*)$ increases with h ($0 \rightarrow 1.88$): the price of smoothing (Figure 4).

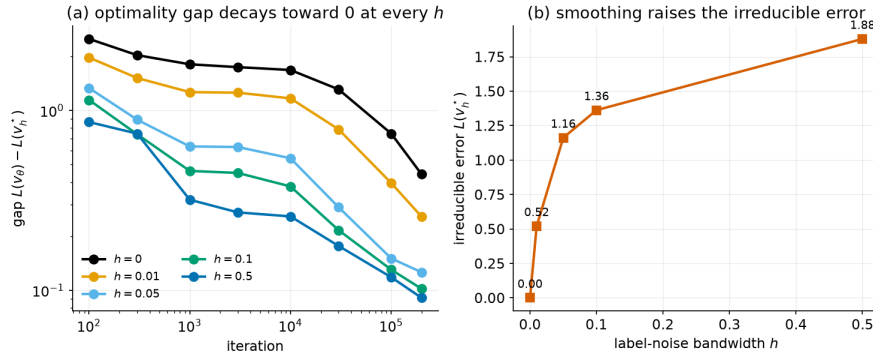


Figure 4: Optimality gap $L(v_\theta) - L(v_h^*)$ decays monotonically toward 0 at every h (left)—it is still positive at the last checkpoint (0.09–0.44), which is the optimisation gap of Section 5; smoothing raises the irreducible error $L(v_h^*)$ (right).

(ii) *Lipschitz floor*. The empirical $\text{Lip}_x v_\theta$ grows from 7.1 at $t=0.5$ to 65.7 at $t=0.99$; the Corollary 4 floor $d/(3L) \approx 0.010$ is $\sim 50\times$ below the observed plateau 0.48, so this particular representation-capacity argument does *not* explain the plateau: the measured plateau is inconsistent with the Lipschitz-floor explanation tested here (we have ruled out *this* mechanism, not every possible capacity/representation effect) and, as the learning-rate-schedule result of Section 4.1 shows, is strongly schedule-sensitive—suggesting an optimisation rather than a Lipschitz-capacity limitation, though we do not have a positive quantitative theory of the optimisation gap itself (Section 5) (Figure 5).

(iii) *Posterior distance*. The MMD to the analytic posterior $\mathcal{N}(\mu_{\text{post}}, \Sigma_{\text{post}})$ falls with h ($0.256 \rightarrow 0.012$) but never reaches 0, and the Sinkhorn distance plateaus near 2.7 (Figure 6). An earlier version of this paper read that plateau as a numerical witness of the floor (5). That reading is wrong, and

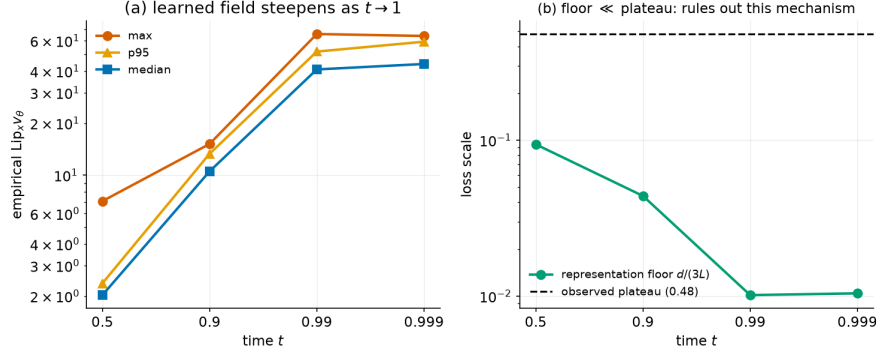


Figure 5: The empirical Lipschitz constant of v_θ blows up as $t \rightarrow 1$ (left); the representation floor $d/(3L)$ is $\sim 50\times$ below the observed loss plateau (right), ruling out this particular Lipschitz-capacity explanation and pointing toward an optimisation-limited plateau instead.

Section 4.2 now reports the correction: the floor is directly computable, equals 0.0353 here—3.5% of $\text{tr } \Sigma_{\text{post}}$ —and the population-optimal atomic law already reaches $\text{MMD} = 0.0089$, well below what the trained model attains. The residual MMD is therefore dominated by kernel bias and the optimisation gap, not by atomicity. Atomicity is real (Proposition 6 is exact) but it is not what these two curves measure.

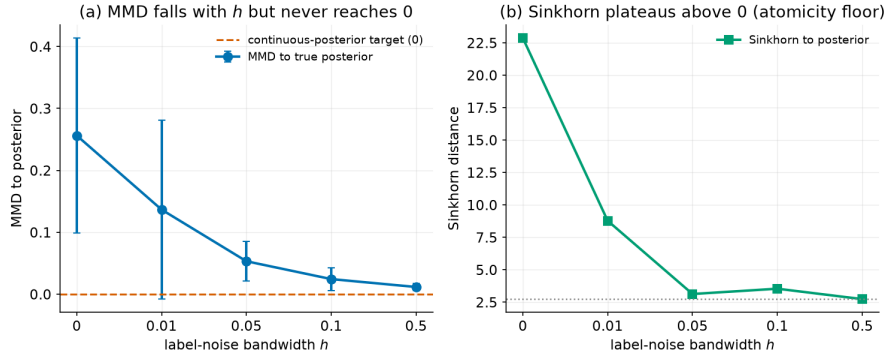


Figure 6: MMD (left) falls monotonically with h but never reaches 0; the Sinkhorn distance (right) drops sharply from $h=0$ and then flattens near 2.7 (it is not monotone over $h \geq 0.05$, varying only within 3.1–3.5 there). Neither approaches 0 at any h —the h -independent atomicity floor of Proposition 6.

(iv) *Does the gap close with budget?* Items (i)–(iii) leave open whether the residual gap at $h > 0$ is an optimisation gap that a longer run would close, or a representation floor that no budget can. We test this directly by extending the $h=0.1$ run to 10^6 iterations under the cosine schedule of Section 4.1, over 5 seeds (Table 9, Figure 7). The covariance reaches the kernel optimum early and stays there ($\text{tr Cov} / \text{tr Cov}_h = 1.65 \rightarrow 1.00$ by 10^5 , then flat to 10^6), while the mean error $\|\text{mean} - \bar{x}_h\|$ keeps falling, $0.188 \rightarrow 0.0534$ at $10^5 \rightarrow 0.0220$ at 10^6 : a further factor 2.4 over the last decade of budget, with no sign of a floor. A representation limit would show as a plateau; none appears within 10^6 iterations. This is positive evidence for the optimisation-paced reading of Section 5, complementing the elimination argument in (ii)—though it bounds the gap from one side only, and does not tell us where the floor eventually lies.

Read this against the *wrong* lens for comparison. The relative velocity error to the collapsed field $(x^i - x)/(1 - t)$ of Section 3.1 barely moves at $h=0.1$ ($0.78 \rightarrow 0.65$ over the same 10^6 iterations) while at $h=0$ it falls $0.78 \rightarrow 0.15$. That is not a representation limit either: the reference v_h^* at $h=0$ does not depend on h , so at $h=0.1$ a model that correctly refuses to collapse *must* stay far from it.

The h -aware Proposition 5 quantities are the only ones that answer the question, which is why they are what Table 9 reports.

iteration	10^2	10^4	10^5	4×10^5	10^6
$\text{tr Cov} / \text{tr Cov}_h$	1.649	1.062	1.002	0.997	1.004
$\ \text{mean} - \bar{x}_h\ $	0.188	0.151	0.0534	0.0314	0.0220
tr Cov	1.091	1.087	1.035	1.018	1.056

Table 9: EXP-1 at $h=0.1$, extended to 10^6 iterations (median over 5 seeds; $\text{tr } \Sigma_{\text{post}} = 1.004$). The covariance matches the kernel optimum from 10^5 on; the mean error is still decreasing at the end of the budget. At $h=0$ under the same schedule and seeds, tr Cov instead falls $1.087 \rightarrow 0.115$ and $n_{\text{eff}} = 1$ throughout—full collapse—whereas here $n_{\text{eff}} = 25.1$ (median; 13.7–43.9 across seeds) and is a property of (X, Y, h) alone, not of training. The two small increases in the mean-error row ($10^5 \rightarrow 2 \times 10^5$ and $4 \times 10^5 \rightarrow 7 \times 10^5$, not shown) are within seed noise. Per seed at 10^6 the mean error is 0.019–0.026; the ratio is 0.99–1.03 for four seeds and 2.85 for seed 4, whose draw of A gives a near-degenerate Cov_h and so an ill-conditioned ratio—the mean error, which does not divide by a small number, is well behaved on that seed too.

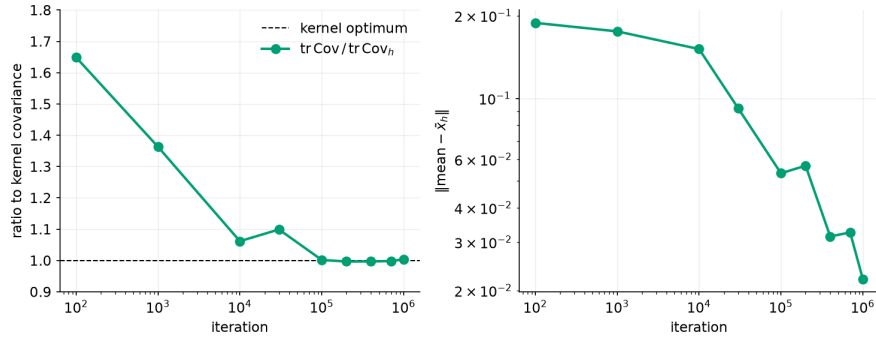
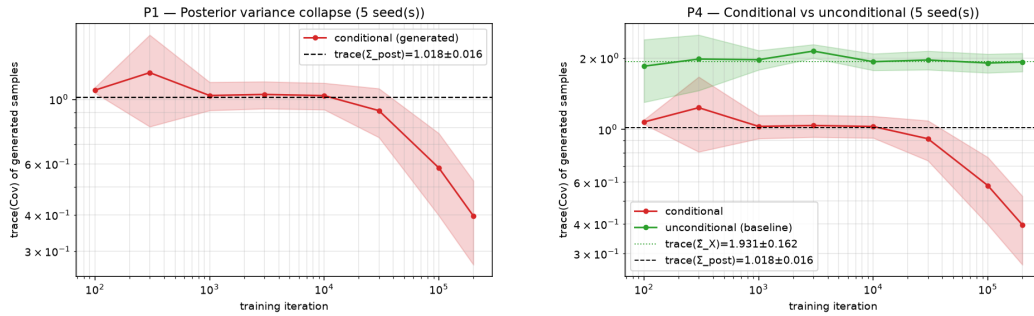


Figure 7: EXP-1 at $h=0.1$ over 10^6 iterations (median of 5 seeds). Left: the covariance ratio to the Proposition 5 optimum settles at 1 by 10^5 . Right: the distance from the generated mean to $\bar{x}_h$ keeps decreasing to the end of the budget—no representation floor is visible.

C.10 SUPPORTING FIGURES AND TABLES



(a) P1: conditional variance collapse.

(b) P4: conditional vs. unconditional.

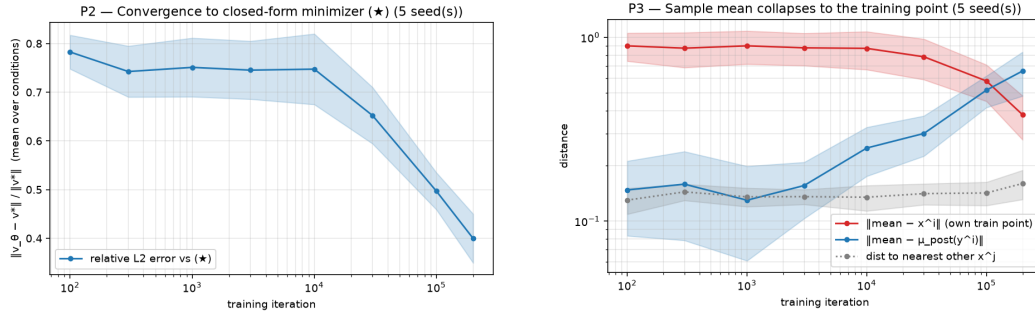
Figure 8: Overtraining collapses the conditional generated variance toward zero, while the unconditional model retains data-scale variance (Corollary 1).

This subsection collects the floats referenced from Section 4. Table 10 gives the full P7 kernel reference and Table 11 the velocity-field match, with the sample-space verification in Figure 15. For EXP-1, Figure 9 plots the P2/P3 curves that descend in lockstep with P1, Figure 10 shows

the $d=2$ endpoint in sample space, Figure 11 the extended cosine-schedule run, and Figure 12 the memorisation ratio; Figures 13–14 cover the adversarial pairing and Table 12 the interpolant-noise contrast. For EXP-2/EXP-3, Figure 16 gives the mode-coverage and MMD curves, Figure 17 the qualitative GMM collapse, Figure 18 the quantitative MNIST curves, and Figures 19–20 the MNIST and CIFAR-10 completion grids.

h	Cov_h (Thm 1)	Σ_{post}	generated tr Cov	ratio-to-kernel	ratio-to-post	$n_{\text{eff}}/200$
0.01	0.583	1.018	0.673 ± 0.187	$1.308 \pm 0.376^\dagger$	0.66	3.5
0.05	0.921	1.018	0.926 ± 0.169	1.002 ± 0.050	0.91	13.6
0.10	1.004	1.018	1.001 ± 0.173	0.994 ± 0.039	0.98	26.0
0.50	1.281	1.018	1.294 ± 0.166	1.011 ± 0.022	1.27	101.5

Table 10: P7 kernel reference, 5 seeds. For every $h \geq 0.05$ the generated variance tracks the exact kernel target Cov_h to within seed noise (ratio-to-kernel ≈ 1.00 , $\text{std} \leq 0.05$)—a verification of the *endpoint law*, stronger than a velocity-field check. † At $h=0.01$ ($n_{\text{eff}} \approx 3.5$, single-atom regime) Cov_h is near-degenerate for many conditions, so the per-condition ratio is unstable and high; the ratio of mean traces $0.673/0.583=1.15$ is closer to 1.



(a) P2: velocity $\rightarrow (\star)$.

(b) P3: mean $\rightarrow x^i$, away from μ_{post} .

Figure 9: The learned velocity approaches the collapsed closed form $(\star)$ (a) and the generated mean migrates from the Bayes posterior mean to the memorised training point (b). Both descend in lockstep with P1. Panel (b) also plots the distance from x^i to the nearest *other* training point (dotted, ≈ 0.17) as a scale reference: at 2×10^5 iterations $\|\text{mean} - x^i\| = 0.38$ still exceeds it, so the mean has not yet resolved x^i from its neighbours at this budget—it does so in the extended run (0.137 at 10^6), and the per-sample memorisation ratio of Figure 12 makes the same point without relying on this scale.

h	rel. error vs. kernel (3)	rel. error vs. $(\star)$	TV of atom mixture
0.01	0.318 ± 0.030	0.599 ± 0.104	0.31 ± 0.18
0.05	0.195 ± 0.019	0.683 ± 0.103	0.22 ± 0.02
0.10	0.162 ± 0.021	0.703 ± 0.092	0.16 ± 0.04
0.50	0.164 ± 0.025	0.791 ± 0.070	0.16 ± 0.02

Table 11: Velocity-field match (5 seeds). v_θ is $2\text{--}4\times$ closer to the kernel field than to the collapsed field at every h . Assigning generated samples to the nearest atom recovers the weights $p_i^{(h)} \propto K_h$ with total variation $0.16\text{--}0.31$. At $h=0.5$ the generated variance 1.29 *overshoots* $\Sigma_{\text{post}}=1.02$, matching the $+h^2\|J\|_F^2$ term of Proposition 8 ($\|J\|_F^2 = 0.4031$, exact).

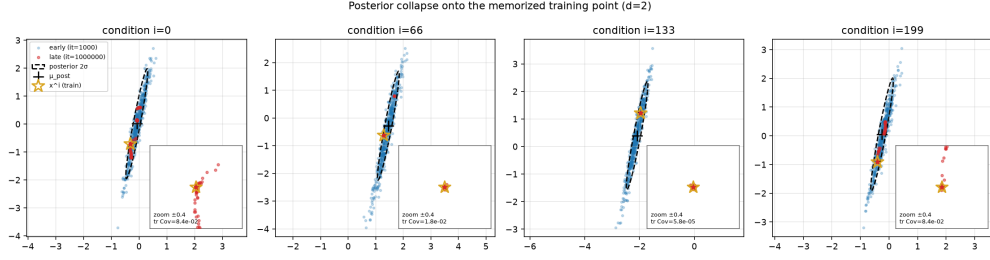


Figure 10: Qualitative $d=2$ collapse (seed 0, cosine-schedule run at 1×10^6 iterations): the sample cloud that initially fills the posterior ellipse (blue, early, 10^3 iterations) contracts onto the memorised training point (red, late) for four conditions y^i . Because the late cloud is one to four orders of magnitude tighter than the posterior scale, each panel carries an inset zoomed to a fixed ± 0.4 window around x^i , annotated with that condition’s tr Cov. The collapse is not uniform across conditions at this budget— $i=133$ is essentially a delta (tr Cov = 5.8×10^{-5}) while $i=0$ and $i=199$ retain a visible residual spread (8.4×10^{-2} , still $\sim 12\times$ below tr Σ_{post})—which is the optimisation-paced behaviour of Section 5, not a failure of Proposition 2.

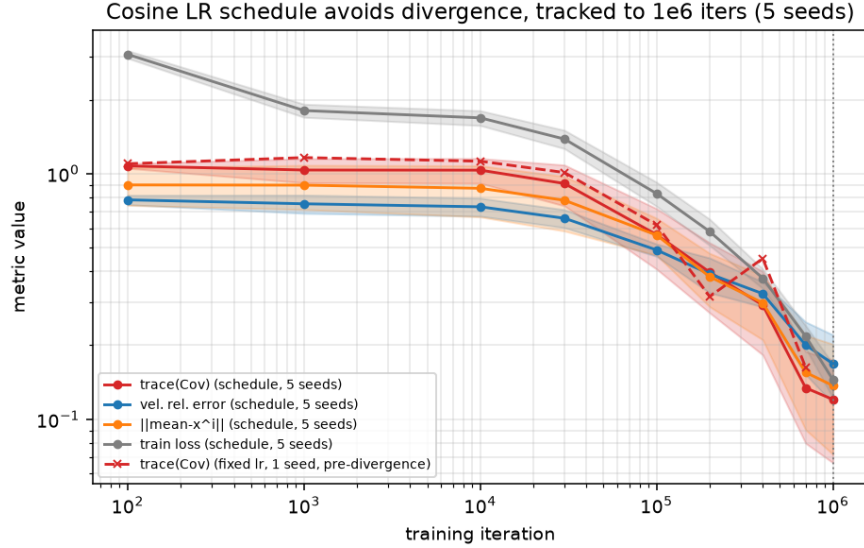


Figure 11: Extended training with a cosine learning-rate schedule (5 seeds, error bars over seeds) tracks all the way to 10^6 iterations without ever diverging, unlike the fixed-lr single seed (dashed), which is cut off at its last pre-divergence checkpoint. On the same seed, the schedule also reaches a lower tr Cov than the fixed-lr run at the matching 7×10^5 checkpoint (0.128 vs. 0.162); the 5-seed mean at 1×10^6 carries enough between-seed spread that it should be read mainly as evidence of stability, not as a precise depth estimate (see main text). tr Cov, velocity error, $\|\text{mean} - x^i\|$, and the training loss all decrease together throughout, identifying P1–P3 as one phenomenon driven by loss $\rightarrow 0$ (the exact minimiser has loss 0).

interpolant σ	0.1	0.3
tr Cov, old target $X_1 - X_0$ (2 seeds)	0.372 ± 0.140	0.429 ± 0.121
tr Cov, corrected target (8) (5 seeds)	0.367 ± 0.146	0.383 ± 0.131
$\ \text{mean} - \mu_{\text{post}}\ $, corrected target (5 seeds)	0.649	0.646

Table 12: Interpolant-noise contrast. Correcting the target leaves the result unchanged within seed noise—a confirmation of Proposition 9(c), not a failure.

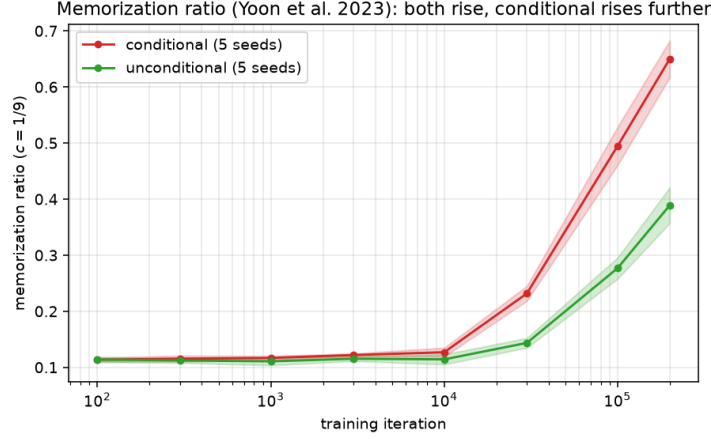


Figure 12: Memorisation ratio (5 seeds each) for the same conditional and unconditional EXP-1 runs as Figure 8. Both rise from the common ≈ 0.11 baseline of a well-calibrated sampler; the conditional ratio rises higher (0.650 ± 0.033 at 2×10^5) than the unconditional one (0.389 ± 0.032), consistent with single-atom vs. full-empirical-measure memorisation (Corollary 1).

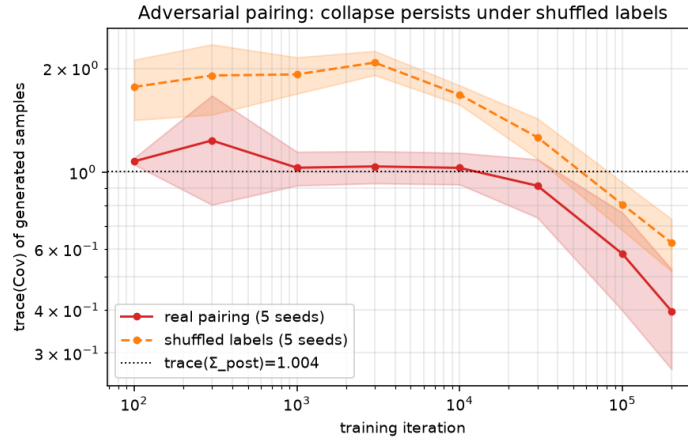


Figure 13: tr Cov under the real EXP-1 pairing (5 seeds, as in Fig. 8) vs. the same architecture and schedule trained on Y randomly permuted relative to X (5 seeds). Both collapse monotonically well below $\text{tr } \Sigma_{\text{post}}$; the shuffled run lags the real pairing at 2×10^5 iterations (0.625 ± 0.108 vs. 0.396 ± 0.127)—an optimisation-gap difference, not a failure of Proposition 2, whose population target is 0 either way.

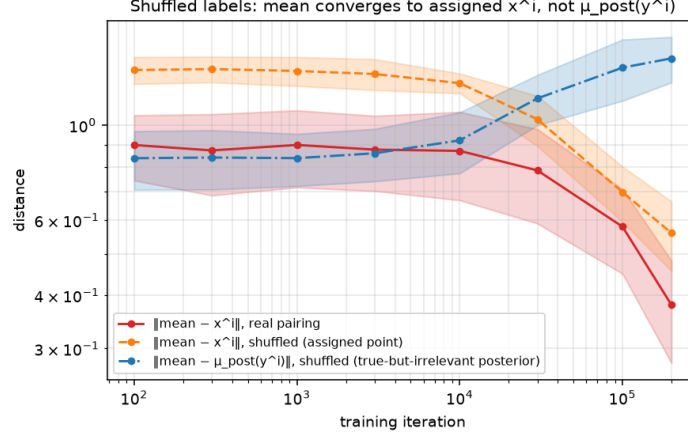


Figure 14: Under shuffled labels, the generated mean converges to the *assigned* training point x^i ($\|\text{mean} - x^i\| : 1.35 \pm 0.10 \rightarrow 0.559 \pm 0.104$) while moving *away* from the true analytic posterior mean of that y^i ($\|\text{mean} - \mu_{\text{post}}(y^i)\| : 0.84 \pm 0.13 \rightarrow 1.437 \pm 0.176$)—the two curves cross between 10^4 and 3×10^4 iterations. Under the real pairing the corresponding gap is far smaller ($\|\text{mean} - \mu_{\text{post}}\| = 0.656 \pm 0.177$ at 2×10^5 , Table 1), since there x^i and $\mu_{\text{post}}(y^i)$ are close by construction.

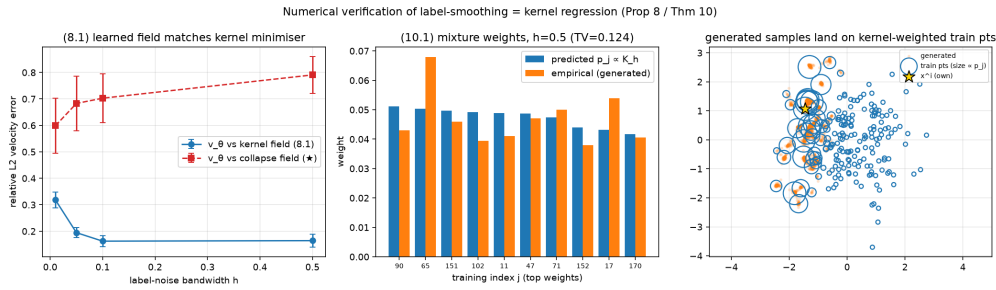


Figure 15: Kernel-theory verification: the learned field matches the kernel minimiser (3) (left); the predicted atom weights $p_i^{(h)} \propto K_h$ match the empirical nearest-atom assignment (centre); and, for one condition y , generated samples (orange) land on the training points weighted most heavily by the kernel (blue circles, radius $\propto p_i^{(h)}$), confirming Theorem 1’s mixture-over-atoms endpoint law directly in sample space (right). The right panel is at $h=0.5$, where the query’s own atom x^i (gold star) is only one of $n_{\text{eff}} \approx 102$ contributors and therefore carries no special weight—the opposite of the $h \rightarrow 0$ regime of Corollary 2.

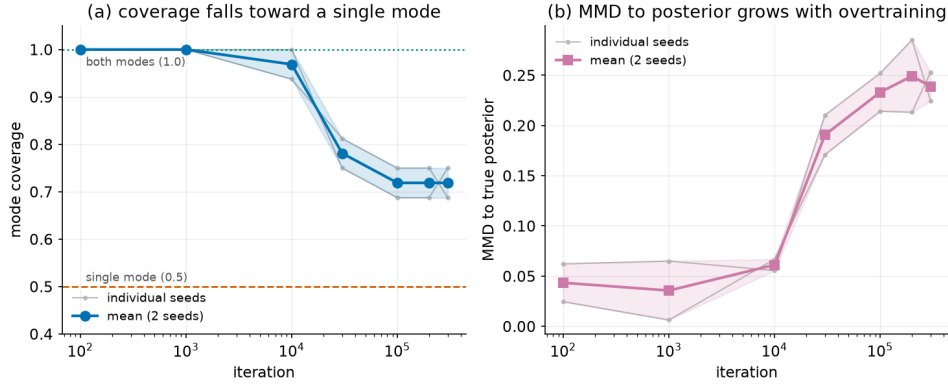


Figure 16: EXP-2 (GMM), mean of 2 seeds with per-seed traces and a $\pm$ std band. **(a)** Mode coverage holds at 1.0 (both posterior modes) until $\sim 10^4$ iterations, then falls toward the single-mode line 0.5. **(b)** The MMD to the true bimodal posterior is near zero in the calibrated phase and grows as overtraining concentrates the samples on one mode. Onset matches EXP-1.

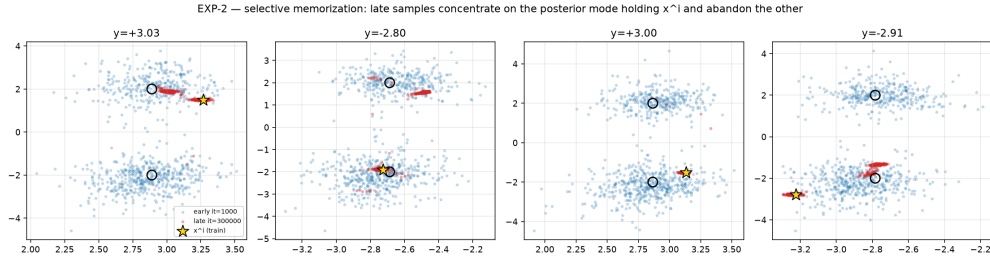


Figure 17: EXP-2 qualitative collapse (seed 0, 3×10^5 iterations). Early samples (blue) cover *both* modes of the bimodal analytic posterior; late samples (red) concentrate on the mode containing the memorised x^i and abandon the other. Assigning late samples to their nearest posterior mode, the share landing in the x^i -containing mode is 0.999, 0.766, 0.990 and 1.000 for the four conditions shown (versus ≈ 0.5 each at iteration 10^3). The one condition that retains 23% of its mass on the abandoned mode shows that this is an optimisation-paced process (Section 5), consistent with the aggregate mode coverage of Section 4.3 settling at 0.72 rather than at the single-mode value 0.5.

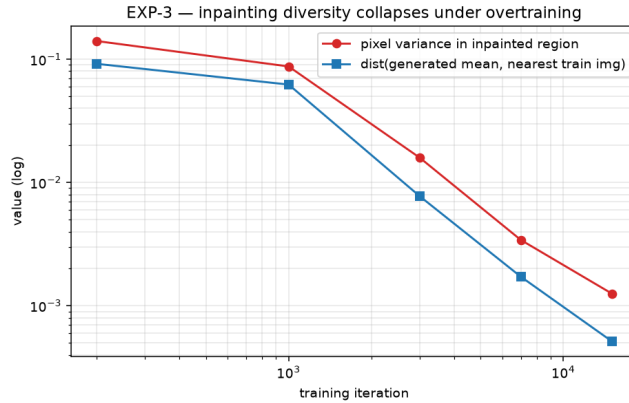
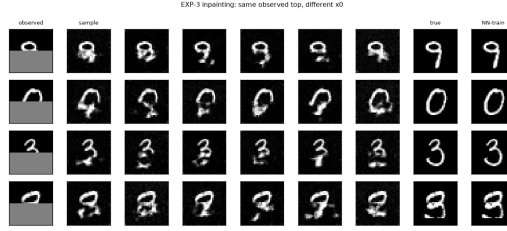
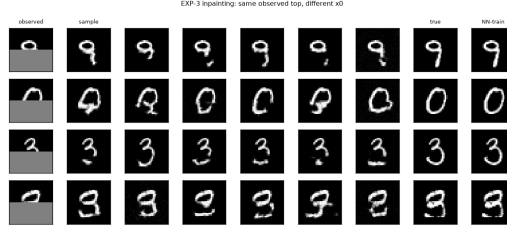


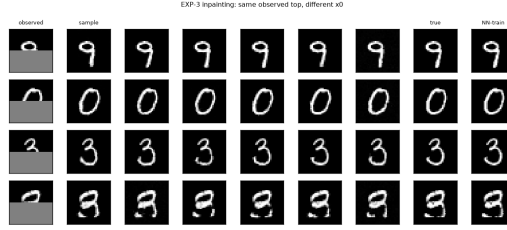
Figure 18: EXP-3 (MNIST inpainting), quantitative collapse, shown for seed 0 (whose checkpoint schedule is the finest of the three; the 3-seed statistics at the shared 15000-iteration checkpoint are in the text). The inpaint-region pixel variance and the distance to the nearest training image both fall by roughly two orders of magnitude as the loss descends.



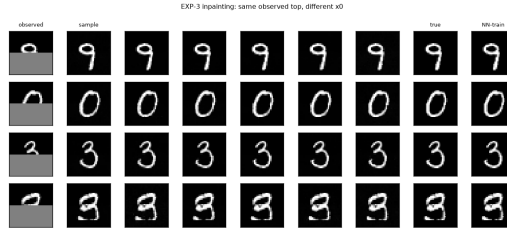
(a) iter 200



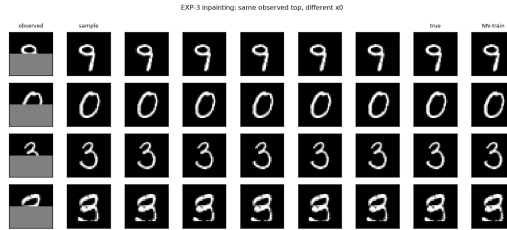
(b) iter 1000



(c) iter 3000



(d) iter 7000



(e) iter 15000

Figure 19: EXP-3 collapse trajectory (MNIST inpainting). Each grid shows completions of a fixed masked digit from different source samples x_0 . Early in training (iter 200) the completions are diverse—a calibrated conditional sampler; as the loss descends they sharpen progressively (iter 1000–7000) and, by iter 15000, are identical to one another and equal to the nearest training image (inpaint-region pixel variance $0.137 \rightarrow 0.0012$ averaged over 3 seeds, a $\sim 100\times$ drop). This is the same loss $\rightarrow 0$ -paced collapse as EXP-1, on real image data, and it reproduces on CIFAR-10 too (Figure 20).



Figure 20: EXP-3 on CIFAR-10 (one seed), same protocol as Figure 19. At iteration 200, completions of the masked bottom half are diverse and noisy; by iteration 15000 they are visually identical to each other and closely match the nearest training image (rightmost two columns), confirming the collapse mechanism is not specific to MNIST’s simplicity. The residual mismatch to that image is larger here than on MNIST (nearest-image distance 0.0040 vs. 0.00077), consistent with CIFAR-10’s harder loss landscape at the same 15000-iteration budget.